Let Us Brew an AI Model

What Tiny Language Models Can Actually Learn When Given One Job

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Zhiqi Research

Building a small language model is relatively easy. Determining what it actually learned is the hard part.

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Preface

This book is a controlled empirical investigation into what a small language model can actually learn when it is given exactly one narrowly defined, objectively testable job.

The task: read a natural-language programming request and produce a Python program, which is then executed against hidden tests.

Along the way we build a laboratory: a controlled world (a sandbox), a teacher (a semantic data generator), a judge (an execution-based evaluator), and rivals (simple baselines). The model is the instrument. The experiment is the subject. And the honest territory includes the moments the model falls flat — and the moment we discovered that our own measuring stick was broken, not the model.

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PART I.

The Question

CHAPTER 1.

Let Us Brew an AI Model

A tiny model is given one job. It reads a sentence — *"write a function that takes two numbers, a and b, and returns their sum"* — and it writes back the right Python:

```
[] def solve(a, b): return a + b
```

We run the hidden tests. It passes. If this were a demo, we'd move on.

But this is not a demo. So before we celebrate, we ask the one question that turns this whole book into an experiment rather than a party trick:

Why did it get it right?

Maybe it genuinely learned that "sum" and "a and b" mean "add them together." Maybe it memorized that exact sentence from training. Maybe it noticed this was nearly identical to an example it saw and copied the answer. Maybe a much dumber machine — one that does nothing but look up the closest training example — could have done the same thing. Or maybe our test wasn't as hard as we thought.

Making the model answer is easy. **Knowing why it answered is the whole game**.

1.1. One job, and a way to check it

We give the model exactly one, narrow, checkable job:

read a plain-English programming request → produce a working Python program

Then we *run* that program against hidden tests. That single choice — executable output plus hidden tests — is what makes this a real experiment instead of a showcase, because it hands us the things a language-model toy usually lacks:

- inputs we control,
- code we can actually execute,
- an objective pass/fail,
- data we can generate in any quantity,
- difficulty we can dial up or down,
- tests we can keep hidden from the model,
- splits we can design to expose memorization,
- audits we can run to catch shortcuts.

1.2. What this book is (and isn't)

This is not a transformer tutorial. It is not a guide to chatbots, agents, or coding assistants. It is a careful, honest investigation of a single question:

> **How much structured programming capability can a small language model actually learn — and how do we tell whether it *learned* it, rather than memorized it or found a cheap shortcut?**

1.3. The part that's actually hard

The natural guess is that *building* a small model is the hard part. It isn't. Once the laboratory works, a ~0.9 million-parameter model will reliably turn a handful of programming sentences into correct Python — in-distribution, that is nearly easy.

The hard part is everything after: proving it didn't just copy, checking whether it still works when the *words* change but not the *computation*, testing whether it can *combine* two things it knows into a third thing it's never seen, and auditing our own evaluator to make sure the number it reports means anything.

This book builds a laboratory, runs those tests, and reports — plainly — what happened. Including the moments the model fell flat, and the moment we discovered our own measuring stick was broken, not the model.

The model was the instrument. The experiment was the subject. Let's brew.

CHAPTER 2.

What Does It Mean to Learn?

We're going to use a word a lot in this book: *learned*. Before we do, we have to be honest about what it means — because "the model learned to add" is a sentence people say all the time, and it's usually hiding a big assumption.

2.1. A simple way to think about it

Imagine a child who aces a worksheet of addition problems. Every problem is like "4 + 3 = ?" and the child gets them all right. We'd be tempted to say "the child can add."

But there are two very different ways to get that perfect score:

1. The child *understands addition* — show them "37 + 58" and they solve it, because they know how to add any two numbers.
2. The child *memorized the exact answers* — they've seen these exact problems before and just remember that "4 + 3 = 7," but hand them a problem they've never seen and they freeze.

A perfect worksheet grade can't tell these two apart. **The worksheet is a bad test**. Only a *new* problem can.

This is the entire premise of the book in one image. Our model gets a perfect-ish worksheet sometimes. The question is always: would it pass the *new* problem?

2.2. What "learned" means here

So we define it narrowly and honestly. A model shows **learned capability** when, to the degree the evidence supports:

1. **It gets better as it trains** — the loss falls, and the thing we care about reliably improves.
2. **It works on examples it never saw** — not just the ones in training.
3. **It survives harmless changes** — different variable names, different wording, a different way of saying the same instruction.
4. **It can combine things it knows** — put two learned pieces together in a new arrangement, not just repeat an arrangement it already saw.
5. **Simpler explanations have been ruled out**. The mundane alternatives — memorization, copying a near-duplicate, a template, a leaked test, a buggy evaluator — must be tested and rejected.

We are deliberately **not** asking whether the model "really understands" in a philosophical sense. We're asking narrower, answerable things: does it generalize? to what? and could a dumber mechanism explain the result?

2.3. The four levels of generalization

This is the book's organizing idea. It is a ladder of how much we should trust a result — from "probably just remembered" to "genuinely combined."

Level 1 — Memorization. The model reproduces something it saw in training: an exact example, an exact constant, a familiar phrase, a near-duplicate. *Question: could this just be remembering?*

Level 2 — Same task, new examples. The model works on a new instance of the same kind of problem. *Question: does it work on problems it hasn't seen before?*

Level 3 — Same problem, different surface. The computation is identical, but the *outer layer* changes: new numbers, new variable names, reworded instruction, harmless distractors. *Question: does it survive changes that shouldn't matter?*

Level 4 — New combination. The model must combine pieces it knows into something that never appeared in training. It learned A, B, and C separately — can it do A *sandwiched inside* B, or A followed by C? *Question: can it put things together, or only repeat what it already saw?*

A rule we'll follow strictly: never mash these four into one number. A model that scores 90% on familiar problems and 0% on a reworded version has not "generalized" — it memorized the phrasing. One number would hide exactly the thing we care about.

2.4. The recurring question: "how did it cheat?"

Because we refuse to trust a single score, every experiment asks the same skeptical question:

How could the model have gotten this without learning what we think it learned?

The usual suspects are always on the table: memorized templates, memorized constants, memorized name patterns, wording quirks, unbalanced data, train/test leakage, near-duplicates, retrieval, a weak evaluator, hardcoded answers, hidden-test leakage.

Every major experiment therefore ends with a **shortcut audit**, six steps: (1) the suspected cheat, (2) why it could explain the result, (3) a control test, (4) what the control found, (5) what that rules out, and (6) what's still unresolved.

That audit — not the headline number — is the book's signature. It's the difference between "the model works" and "we know why the model works."

CHAPTER 3.

Give the Model One Job

Before we can run any experiment, we have to pin down the task so precisely that a machine can carry it out and we can score it objectively. Vague tasks give vague answers; a task with concrete inputs, runnable outputs, and hidden tests gives crisp, checkable answers.

A well-designed exam question is the model. It's not enough to say "write a program about numbers." We need: exactly what the input is, exactly what the program should do, and exactly how to check if it's right. That's what we're about to define.

3.1. The task

The model receives a **natural-language specification** of a programming task and must produce a **Python program** that solves it. That program is then executed against hidden tests.

Concretely, the model sees something like:

> "Write a function that takes two numbers, a and b, and returns their sum."

and must produce:

```
[] def solve(a, b): return a + b
```

We score it by running that program against a set of hidden test cases (inputs with expected outputs) and checking whether the program's output matches, semantically, for every case.

3.2. The task schema

Every task is described by a small, formal schema — the ground truth that drives everything. The key insight is that **one semantic record generates all three** things we need: the reference program,

the natural-language spec, and the hidden tests. That way there is no drift between "what the task is" and "how we ask for it" and "how we test it."

The semantic core captures the operation and its arguments:

> [] { "operation": "add", "args": [{"name": "a", "type": "int"}, {"name": "b", "type": "int"}], "output_type": "int" }

From that single record, the teacher generates: 1. the reference Python (`def solve(a,b): return a+b`), 2. a natural-language specification (in one of several phrasings), 3. a set of hidden tests (ordinary, zero, negative, boundary, large cases).

3.3. The pure-function constraint

For the pilot we constrain the generated programs to be **pure functions**. No side effects, no file I/O, no randomness. This makes evaluation deterministic and unambiguous: for a given input, there is one correct output. It is a deliberate simplification, and it is exactly what makes the task cleanly measurable.

3.4. Hidden tests

The hidden tests are the heart of evaluation. Each task carries an independent set of test cases designed to catch real errors, not just a single happy path. For an arithmetic task we include ordinary values, zero, negatives, boundary values, and large-magnitude values. The tests are generated independently of the reference program's *logic* — computed from the semantics, not copied from the solution — so that a model that merely memorizes the reference still can't predict the exact test suite.

3.5. The success criterion

For a single problem, the success criterion is:

> **Problem-level hidden-test pass@1** — did the one program the model produced pass *every* hidden test for that problem, under deterministic/greedy decoding?

This is deliberately strict. A program that passes 7 of 8 hidden tests *fails* the problem. That strictness is what keeps the measurement honest: we are not measuring "close," we are measuring "correct."

3.6. Why this specific task

Natural-language-to-Python is a good laboratory task precisely because it has everything a controlled experiment needs:

- **Deterministic**. Same spec, same program, same tests.
- **Executable**. We run real Python and compare real outputs.
- **Objectively scorable**. Pass/fail, no judge-subjectivity.
- **Synthetic**. We can generate as much controlled data as we want.
- **Controllable difficulty**. We can escalate from `a+b` to compounded operations.
- **Compositional splits**. We can train on pieces and test on unseen combinations.
- **Memorization tests**. We can plant canaries and check for exact reproduction.
- **Auditable**. We can inspect the exact generated program and run shortcut audits.

The model could not ask for a clearer job. Now we have to build the tools that let us run and score it: a controlled world, a teacher, a judge, and rivals.

PART II.

Build the Laboratory

CHAPTER 4.

Build a World We Can Control

A model that answers natural-language questions is easy to build and hard to evaluate. But there is one class of task where the evaluation is not a judgment call at all: **write a program**. We write a description, the model writes code, and a computer can tell us, with certainty, whether that code does what was asked. There are no opinions about the answer, no ambiguous grading rubric, no "well, it's sort of correct." Either the program returns the right thing for every test, or it does not.

That one observation is why this whole book rests on the `natural-language → Python program` task. It gives us something few machine-learning setups have: *objective, executable, reproducible* evaluation. But the task alone is not enough. Before we can run any experiment, we need a world the model can live in — a world that is safe, that is *deterministic*, and that is small enough that we can reason about everything that happens inside it.

4.1. Why we need a world at all

Imagine hiring someone to write code, but not fully trusting them. We would not hand them the keys to the server. We would give them a sandbox — a walled-off room where they can work, but where the dangerous tools are simply not present.

That is exactly what we do with the model. It writes Python, and we need to run that Python safely and deterministically, while deciding precisely what it is *allowed* to do. The model could, in principle, write a program that reads files, connects to a network, or deletes things. It has no reason to — but we are not relying on its good intentions. We are building a world where those actions are *impossible*, not just discouraged.

The deeper reason is that a controlled world makes the *experiment* clean. If the model's program could do anything — read a file, check the time, generate a random number — then two runs of the

same program could give different answers, and we could never attribute a failure to "the model didn't learn this" versus "the program did something nondeterministic." We want the world to be so constrained that **the only source of variability is the model itself**.

4.2. The principle: whitelist, not blocklist

The naive way to constrain a program is a blacklist: deny `import`, deny `eval`, deny `open`. This is the instinct most people reach for first. But a blacklist is fundamentally fragile — there is always some construct we forgot. Python has an enormous surface area, and attackers (or, in our case, just curious models) find the gaps. Denying a hundred things is not safety; it is a constant game of catch-up.

The robust way is the opposite: **an allowlist**. We parse the program into its abstract syntax tree — the structured representation of the code, before it runs — and walk through every node. If any node type is not *explicitly permitted*, we **reject the entire program**. There is no "mostly allowed." If the model emits even one disallowed construct, the whole thing is refused before it ever executes.

The difference is between a list of banned words and a grammar. A banned-word list can be evaded by creative phrasing. A grammar says: these are the only forms that exist. Anything outside the grammar is not merely suspect; it is *unparseable in our world.* That is the philosophy of the sandbox.

This is the single most important design decision in the whole apparatus, and it deserves to be stated plainly: **the sandbox does not try to catch bad programs. It states upfront what is allowed, and everything else is refused.**

4.3. The capability spec

For the pilot, we whitelist a deliberately small subset of Python — small enough that we can hold it in our heads, large enough to express real programming:

- literals (integers, floats), variables, assignment, `return`
- arithmetic: `+ - * / // % **`
- comparisons, boolean operators
- `if / elif / else`, `for`, `while`, `break`, `continue`, `pass`
- lists, indexing, slicing
- a small set of built-ins: `abs, min, max, sum, len, range, int, float, round`
- simple `def` functions

Everything else is rejected: imports, file I/O, network, subprocess, `eval`/`exec`, `class`, `try/except`, arbitrary attribute access, comprehensions, lambdas. We do not include these because they are dangerous; we exclude them because they are not needed for the task we are testing, and because each one we add makes the world bigger and harder to reason about. A smaller

world is a *better* world for this book, because every construct that is allowed is one we must understand the model's relationship to.

We also version this spec (`python_capability_v0_1`) so that later, when we give the model *more* Python (Chapter 18), we can point to exactly what grew. Versioning the world is what lets us claim, later, that a change in behavior was caused by a change in the world, and not by something else that drifted.

4.4. Resource and determinism policy

Beyond what is allowed syntactically, we also control *how* the program may run. This is the second half of making the world trustworthy:

- maximum AST depth and node count (a program can only be so complex),
- maximum nesting,
- a hard execution timeout (a program that loops forever is a failed program),
- recursion protection,
- float-comparison tolerance (because `0.1 + 0.2` is not exactly `0.3`),
- **no randomness, no time, no environment access**.

That last point is the quiet heart of the whole design. A program with the same input must always produce the same output. If it does not, the experiment is contaminated. We cannot tell whether a failure means "the model didn't learn this" or "the program rolled the dice." By making the world deterministic, we eliminate that ambiguity entirely. The book's every later claim rests on this.

4.5. The pipeline

```
source -> parse -> AST -> whitelist validation -> compile
        -> restricted execution -> semantic comparison
```

Validation happens on the tree *before* execution, so disallowed code never runs. Execution uses a stripped set of globals that exposes only the allowed built-ins — there is no `__import__`, no `open`, no `eval`. The program can only see the tools we chose to give it.

4.6. An honest caveat

It is important to be clear about what this sandbox is *not*. It is a **teaching and evaluation** sandbox — deliberately strict about the capability subset, but not a claim of production-grade adversarial isolation. We are not trying to secure a server against a hostile attacker; we are trying to run generated code and measure whether it does what we asked. An attacker with time and motivation could

likely find ways around a small AST allowlist. We state this limitation plainly rather than overstating what the sandbox guarantees, because the book's credibility depends on honest boundaries, not on impressive-sounding claims.

4.7. Testing the world

The sandbox is the foundation of trust in every later result. If the sandbox leaked — allowed something disallowed, or failed to be deterministic — every score in the book would be suspect. So the very first thing we do is not run any experiment at all. We **test the world itself**.

We build a conformance suite of 31 cases: programs that use each allowed construct and must pass, plus programs that use each disallowed construct (`import`, `eval`, `class`, `try`, comprehensions, lambdas, arbitrary attribute access) and must all be rejected. A sandbox that only permits things is easy; a sandbox that also *correctly refuses* everything outside its grammar is the real achievement — and that is what we verify before we trust a single number it produces.

This is the first instance of a discipline the book will return to again and again (and which we will see was worth it, painfully, in Chapter 10): **before we use a tool to measure something, we verify that the tool measures what we think it does**. Here, the tool is the world, and we refuse to run a single experiment until it passes its own examination.

CHAPTER 5.

Build the Teacher

We need data. But not just any data — we need data we can *trust* to be fair, and data we can carve up to test the model honestly.

A teacher preparing worksheets for a test is the model. A bad teacher writes twenty questions that are all the same question with different numbers — the student can memorize one template. A good teacher writes questions that look different on the surface but test the same idea, so a student who actually learned the idea passes whether or not they've seen the phrasing.

Our **teacher** is that good worksheet-writer. But there's something special about the way we build it, and it's the single most important design decision in the whole book.

5.1. Semantic-first: start with the meaning, not a template

The naive way to generate data is to write a template and fill in a few blanks:

```
template -> sentence -> code
```

We write "add {a} and {b}" and a few code snippets, and we are done. This is fast and easy, and it's exactly the wrong way to do it. The problem is that the sentence and the code are generated **independently**, from a template, so there's no guarantee they mean the same thing. We can end up with a sentence that says "add" but code that subtracts, or a test that checks the wrong thing.

We do the opposite. We start from the **meaning** and generate everything else **from it**:

```
SEMANTIC INTENT
   -> PROGRAM STRUCTURE
      -> REFERENCE CODE
      -> NATURAL LANGUAGE
```

```
-> HIDDEN TESTS
```

The key insight is at the top: **one semantic record drives all three**. There is no way for the sentence to disagree with the code or the tests, because they are all the same meaning, rendered three ways. If we start with "the function should return `a + b`," the sentence says "sum," the code adds, and the tests check addition — and they can't drift apart, because they all came from the same seed.

5.2. What a "semantic record" looks like

The meaning is captured in a small, structured object. It is not English and not code; it is the *idea* underneath both. For an addition task it looks like this:

```
[] { "operation": "add", "args": [{"name": "a", "type": "int", "role": "operand"}, {"name": "b", "type":
"int", "role": "operand"}], "output_type": "int" }
```

From this single object, we generate:

1. **the reference program** — `def solve(a, b): return a + b`,
2. **the natural-language prompt** — e.g. *"Write a function that takes two numbers, a and b, and returns their sum"*,
3. **the hidden tests** — e.g. `{args:[-28,47], expected:19}`, `{args:[0,0], expected:0}`, and edge cases we'll get to.

This is the heart of the teacher. The pump is `make_task`, which takes an operation and a seed, builds the semantics, and then calls three sub-generators that all read that one record.

5.3. The four sub-generators

The teacher is really four small pieces working together, all fed by the same semantics:

1. **`make_semantics`** — builds the semantic intent: the operation plus its argument descriptors. It pulls argument names from a controlled set (so we can vary them later), assigns an input/output type, and tags the task with a complexity score.

2. **`generate_reference`** — turns the semantics into the ground-truth Python. This is the correct program the grader is validated against. For `add` it emits `def solve(a,b): return a+b`; for `list_sum` it emits a loop over the list.

3. **`render_nl`** — turns the same semantics into English, using one of several **template families**. A template family is just a different way of phrasing the same instruction. For addition, we have phrases like:

- *"Write a function that takes two numbers, a and b, and returns their sum."*
- *"Implement solve so that given a and b it returns their sum."*

- *"Define a function solve(a, b) that returns a + b."*

Same meaning, three different wordings. This is what gives the dataset linguistic diversity — and, more importantly, it's what lets us later test whether the model learned the *idea* or just one *sentence*.

4. **`generate_hidden_tests`** — generates test cases **from the semantics**, not from the reference program's logic. It deliberately covers edge cases: ordinary values, zero, negatives, boundary values, large magnitudes, and (where legal) empty lists. For each it computes the expected value from the operation itself.

5.4. Why the tests come from the meaning, not the code

This is a subtle but crucial point. If we generated the tests by looking at the reference program, a model that memorized the reference would already know what the tests check. By generating them from the *semantics* instead, the tests probe whether the model independently understands the operation — whether it can produce the right answer for inputs it could not have seen in the reference.

5.5. Maximizing controlled diversity

A good teacher does not just produce many examples; it produces *diverse* ones while holding the semantics fixed. We deliberately vary three things:

- **constants** — different values each task, so the model can't memorize outputs,
- **identifiers** — `a, b` vs `left, right`, so it can't tie the operation to particular names,
- **wording** — several template families per operation, so it sees the same idea said many ways.

This controlled diversity is what makes the *generalization tests* later possible. Because we control *how* we vary the surface, we can also *hold it out* and test whether the model generalizes. That's not an accident; it's the whole point.

5.6. Avoiding accidental correlation

The danger in any generated dataset is **accidental correlation** — a pattern that makes the answer trivially predictable from the surface. The classic failures:

- every addition example uses the word "total";
- every maximum example uses the variable `x`;
- every multiplication has exactly three arguments;
- every loop example is exactly one sentence long;
- every operation uses one unique wording.

If those correlations exist, a model (or a trivial classifier) can solve the task without learning the computation. The generator is built to break them, and then — this is the crucial part — we **check that it actually did**.

5.7. The leakage audit

Before trusting any dataset, we ask: *can a trivial classifier predict the answer from the surface alone?* The method is deliberately crude: train a TF-IDF + logistic regression model on the natural-language prompts and see how well it predicts the target operation from the words alone. If a dumb classifier succeeds, the wording is leaking the answer.

And it caught us. On our first pilot renderer, a trivial classifier predicted the operation from the wording with **100% accuracy**. That's a red flag, and it's a valuable catch: it means our wording was so tied to the operation that a model which "learned" might just be exploiting the phrasing. We recorded it as a findable artifact (finding-001) and noted exactly what it means for the experiments that follow.

This is the teacher doing its real job. A dataset generator that only produces data is half a tool. One that produces data **and audits it** is a laboratory instrument.

5.8. The takeaway

The teacher is not a bulk-data tool; it is a controlled instrument. Its job is to produce data with *controlled* diversity and *audited* cleanliness, so that when the model succeeds we can attribute it to learning the computation — not to the data accidentally giving away the answer.

And the reason this matters so much is the whole book's premise. Everything later — the generalization tests, the composition test, the capability map — depends on the data being a *fair* test. The teacher is where that fairness is built.

CHAPTER 6.

Build the Judge

How do we know a student's answer is right? We check it against the answer key. But what if the answer key is wrong? Then every student who gets it "right" is actually wrong, and we'd never know.

That is the risk with our **judge** — the evaluator that decides whether a generated program is correct. It is the instrument whose readings the whole book depends on. If the judge is wrong, every number in this book is wrong. So we build it carefully, and then we attack it.

This is the second pillar of the apparatus, and it is in some ways the most important. The world (Chapter 4) decides what the model is *allowed* to do. The teacher (Chapter 5) supplies the material to learn from. But the judge is the thing that decides whether the model *got it right*. A subtle way to frame the whole book: we are not really studying the model. We are studying the model *through* the judge. So the judge's integrity sets an upper bound on the trustworthiness of everything that follows.

6.1. Why an execution-based judge

The naive way to judge a generated program is to compare it, as text, to the expected answer. This is cheap, and it is almost worthless. Two programs that solve the same task can look completely different as strings — one uses a loop, another uses `sum()`, yet both are correct. And two programs that look nearly identical can behave differently. String comparison measures *how the code is written*, not *whether it works*. That is not what we care about.

So the judge does the thing that makes this whole task special: it **runs** the program. It executes the generated code against hidden tests and compares the actual *output* to the expected output. This is what gives us objective, unambiguous judgment. There is no opinion about whether the code is

elegant or the "right" style — either it returns the correct result for every test, or it does not.

6.2. The four-stage pipeline

We score a candidate program through four stages; a later stage only matters if the earlier ones pass. Each stage is a filter, and each exists for a specific reason.

1. **Syntax pass** — does the program parse as valid Python?
2. **AST / capability pass** — is every node within the whitelisted subset? If the program smuggles in an `import`, it is rejected here.
3. **Execution pass** — does it compile and run without error under the restricted environment?
4. **Semantic pass** — for each hidden test, does the program's output match the expected output?

The primary metric is **problem-level hidden-test pass@1**: did the one program produced pass *every* hidden test for that problem?

Why four stages rather than one? Because each stage isolates a *different* kind of failure, and separating them tells us *why* a program failed. A syntax error is a very different problem from a capability violation (the model tried to use something outside its world), which is different again from a runtime error (the code ran but crashed on a test), which is finally different from a semantic error (the code ran but produced the wrong answer). If we collapsed all four into one "it failed," we would lose the ability to diagnose. The judge does not just tell us right or wrong; it tells us *where* things go wrong, and that diagnostic power is worth keeping.

6.3. Semantic comparison: don't trust string equality

A naive evaluator would check `output == expected`. That fails in subtle ways. Two programs can produce the "same" result that isn't string-equal: floats with rounding, a dict with keys in a different order, nested structures. So the judge uses a proper **semantic comparator**:

- integers compared exactly,
- floats compared with an explicit tolerance,
- booleans compared as booleans (not as 0/1),
- lists compared element-by-element, recursively,
- dictionaries compared key-insensitively (order doesn't matter),
- nested structures compared recursively.

The comparator itself is tested — we don't assume it's right just because it's simple. And the tolerance matters: floating point arithmetic is not exact, so `0.1 + 0.2` does not come out to exactly `0.3`. A judge that demanded exact equality on floats would reject correct programs. A judge with no tolerance at all would be as broken as one that accepted anything. Choosing the right tolerance is a real design decision, and it is the kind of thing that quietly shapes every number the judge produces.

6.4. Mutation testing: red-teaming the judge

The most important step is treating the judge as something that must be *proven* sound. We **mutate** the reference solution into deliberately-broken versions and ask: *does the judge catch them?*

We generate mutants such as: `+` ⟶ `-`, `>` ⟶ `>=`, remove the `return`, swap the arguments, off-by-one a loop boundary, use the wrong variable. Then we run each mutant through the hidden tests. A good test suite **rejects** an obviously-wrong mutant. A weak test suite lets it pass.

The mutation score — the fraction of wrong mutants the tests catch — is a measure of how much we can trust the judge. If an obviously-broken program passes our tests, the tests are too weak, and we add more until they discriminate.

This is the "test the tester" principle, and it is the reason we do not trust the judge just because we wrote it. The judge is the instrument of measurement, so we deliberately give it wrong inputs and check that it rejects them. It is a rare and valuable discipline: most people build a measuring stick and assume it measures. We deliberately try to break ours.

6.5. We caught our own judge being wrong

Here is what makes this chapter honest rather than aspirational: the very first time we evaluated our model, it **scored 0.000** — reported a total syntax failure on every problem. And the model wasn't wrong. The *judge* was.

The root cause was a generation bug on our side: the model produced a complete, correct program, but it was followed by padding (null bytes), and `ast.parse` rejects null bytes. So the judge scored a perfectly valid program as a syntax failure. Once we truncated the padding, the model's true score emerged: **0.897**.

That moment is worth more than any passing test. It is the book's core lesson applied to ourselves: **never trust a number from a system we have not audited**. If we had taken the first 0.000 at face value, we'd have concluded the approach failed. Instead, finding the judge's bug told us the truth — the model was actually working.

What is striking is that this bug was not exotic. It was a small, mundane detail — a sentinel byte used to mark the end of a generated program — colliding with Python's parser. We would never have found it by reasoning about the model. We found it because we looked at the judge's *intermediate* output (the syntax scores) rather than jumping to the headline number, and because we already had the habit of doubting the measurement. That habit, more than any single fix, is what kept the book honest.

6.6. The honest summary

The judge is not a helper; it is the instrument whose readings we stake the book on. So we built it strictly (four stages, semantic comparison, tested comparator), then we red-teamed it (mutation

testing), and then — when it gave a suspicious answer — we attacked it until we found our own bug. Only a judge that survives that scrutiny deserves to be trusted.

And the deeper lesson is one we will keep relearning: in any measurement, the instrument is part of the experiment. The model and the judge are both subjects. We spend almost as much effort verifying the judge as we do training the model — and that is not waste. That is the only way to trust the numbers at all.

CHAPTER 7.

Build the Rivals

A model that scores 90% sounds impressive — until we ask the hard question: *could a much simpler machine do the same thing?* This chapter is about building those simpler machines, our **rivals**, and why they are the most important comparison in the book.

7.1. The principle

The value of a neural model's success is judged *relative to a baseline.* If a hand-written rule or a lookup table solves the task as well as the trained model, then the model isn't demonstrating capability — it's re-deriving something simpler. The baselines are what turn "the model works" into "the model does more than a simpler machine could."

This is a point that deserves to be stated bluntly, because it is easy to overlook: a number in isolation is almost meaningless. "The model solved 90% of the problems" sounds like a triumph. But if a ten-line lookup table also solved 90%, the model's result is not evidence of learning — it is evidence that the task can be solved without learning. The baseline is what gives the number its meaning.

This is the single question the whole book keeps asking:

> **Could a much simpler machine do this?**

And it is remarkable how often the answer is "yes" at first. That is a feature of the investigation, not a defect. Finding that the task is too easy for a simple machine tells us exactly how to make it harder, which is how the whole brew progresses.

7.2. Rival 1 — the template / rule system

Our simplest rival is a deterministic pattern-matcher. It recognizes known keywords in the specification — "sum" ⟶ addition, "maximum" ⟶ max, "length" ⟶ list length — and emits the corresponding predefined code template. It does no learning whatsoever. It is, in the most honest sense, *dumb* — a set of hand-written if-then rules.

This rival answers: *how much of the task is solvable WITHOUT learning?* If a keyword recognizer does fine, then the task's "hard part" is just pattern-matching, and a model learning it isn't learning much.

Building this rival is almost too easy. It is a few rules that map common words to common code. But its simplicity is precisely its value. It is the lowest useful baseline: the floor of what any reasonable person would expect a hard-coded system to achieve. If our learned model cannot beat this floor, then the model is not learning — it is re-deriving a rule table, more expensively and less reliably.

7.3. Rival 2 — retrieval

Our second rival is even more interesting. A **retrieval** system builds a simple TF-IDF index over the training prompts. Given a new task, it finds the most similar training example and *reuses that example's code*.

This rival answers: *can apparent neural competence be explained by retrieval?* If the test set is close enough to the training set, a lookup that does nothing but find the nearest example will do perfectly — and a model that also scores high might just be doing the same thing in a more complicated way.

The retrieval rival is the more insidious of the two, because it is the one that threatens the model's success most directly. A template system shows whether the task is trivially pattern-matching. But a retrieval system probes something subtler: whether a model's score could be explained by *surface similarity to training data*. If a TF-IDF lookup can find a near-duplicate prompt and reuse its answer, then a neural model scoring similarly might be doing exactly that — matching surface form, not understanding. Retrieval is the closest thing we have to a detector for "the model is just copying."

7.4. Why these two matter

These two rivals are the scientific controls. Together they bound what any result means:

- If the **template** baseline is high, the task is pattern-recognizable — the model's win is shallow.
- If the **retrieval** baseline is high, the task is near-duplicate-solvable — the model's win may be memorization or surface similarity.

And the third possibility is the one that matters most: **if the model falls *below* both baselines under some change, we have found a genuine limit of the model's learning.** The rivals are not just

a ceiling to beat; they are a floor that reveals when the neural model is doing something *no simple machine can do* — and, more valuably, when the neural model has become *worse* than a simple machine, which is a powerful signal of surface-tied, non-general learning.

Our experiments produced exactly this signal. On the pilot, retrieval scored **1.000**: a lookup table solved the whole test set. That told us — correctly — that the task was too easy to force generalization, and we had to make it harder. The rivals did their job: they caught that the impressive-sounding model was being tested on a task that didn't require much learning.

Then, in the surface-generalization chapter, the story inverted. Under a *wording* change, the model collapsed to **0.000** while the template system, which recognizes keywords and is indifferent to exact wording, stayed at **0.875**. Here the rivals were not a ceiling — they were a *floor*, and the model fell through it. That was the most informative result of the experiment: it showed that the model's apparent capability was tied to exact surface form in a way that even a dumb rule system was immune to.

7.5. The lesson

A neural model is not impressive merely because it works. It is impressive only when it does something a simpler machine cannot. Every result in this book is therefore interpreted relative to these rivals — and the moments where the model *falls below* them are some of the most informative results of all.

The deep takeaway is that baselines are not a formality or a box to tick. They are the reference frame that gives every measurement its meaning. Without them, a score is just a number. With them, a score becomes a claim we can interrogate. And building rivals that are *deliberately simple* is what makes them a strong test: the simpler the machine that can match or beat the model, the more damning (or the more shallow) the model's achievement turns out to be.

PART III.

The First Brew

CHAPTER 8.

The Smallest Brain

Now the student. Everything so far — the world, the teacher, the judge, the rivals — exists to ask one question about this thing. So let's meet it.

Meet the model: a small, deliberately plain decoder-only transformer. It's not fancy, and that's the point. Just like a chemistry lab uses simple glassware so we can see exactly what's happening, we use a simple, transparent model so we can understand exactly what it's doing. A 20-layer exotic architecture would be a black box; a clean, standard transformer is one we can reason about.

8.1. Why the simplest possible model

The temptation when choosing a model is to reach for the newest, most sophisticated thing — the attention variant that promises to be smarter, the architecture that looks impressive in a diagram. That instinct is understandable, but it fights the entire purpose of a controlled experiment. If the model is complex and mysterious, then when it fails, we cannot tell whether the failure is because of some fundamental limit of learning, or because of a quirk of our exotic architecture.

So we choose the opposite. The student is a **completely standard, off-the-shelf decoder-only transformer** — the same basic architecture that powers the big language models, just scaled way down. Nothing clever, nothing bespoke. This is a deliberate choice, and the reason is not modesty; it is method. A plain, well-understood model means that when something unexpected happens, we can attribute it to *learning* rather than to an architectural mystery. We want the model to be the simplest possible instrument, so that the experiment — not the gadgetry — is the subject.

8.2. What it is

The student is a deliberately simple, transparent **decoder-only transformer**. The goal is not novelty; the goal is that we can understand exactly what it is doing. A 20-layer exotic architecture would be a black box; a clean, standard transformer is one we can reason about.

For the pilot, the configuration is small and honest:

- vocabulary: ~291 tokens (a compact BPE trained on the task corpus),
- hidden dimension: 128,
- layers: 4,
- attention heads: 4,
- context length: 128,
- GELU activation, Pre-LayerNorm, AdamW optimizer.

Each of these is a deliberate choice that we can justify. The context length of 128 tokens, for example, is chosen because a pilot task's prompt-plus-program fits comfortably within it — long enough to hold a full task, short enough that the model is not wasting capacity on irrelevant span. The 4 layers and 128-wide hidden dimension keep it fast to train on the modest hardware available, which matters enormously: if training takes an hour, we iterate on the *experiment*; if it takes a day, we wait on the *model*, and the brew slows to a crawl. A small model is a fast iteration loop, and a fast loop is what lets a scientific investigation move.

8.3. The honest parameter count

A common mistake is to quote just a round "10M parameters." We refuse to hide the important part. The model has **884,224** parameters, of which only **830,592** are the actual computation. The other **53,632** are the **embeddings** — the token embedding ($291 \times 128 = 37{,}248$) plus the positional embedding ($128 \times 128 = 16{,}384$) — which is the part that just stores the vocabulary and position, not the part that does the learning. If we do not separate them, "0.9M parameters" can mislead: a big fraction might just be the embedding tables, not any computation.

So we always report the split: embedding vs. non-embedding.

This matters more than it might seem. At tiny scale, the ratio of vocabulary to computation is high, which means a naive "M parameters" figure can be dominated by a table lookup that does no learning at all. Separating the two lets us and the reader see the *actual* computation the model brings to bear — the part that must do the learning. It is a small discipline, but it keeps the conversation honest.

8.4. The architecture smoke test

Before we train it for real, we run an **architecture smoke test** — a tiny, fast check that the model's forward and backward passes work, that the parameter count is what we expect, and that the ini-

tial loss is sensible (roughly the log of the vocabulary, since a fresh random model's best guess is uniform).

This is the same discipline as the very first smoke test, but one level up: we are not checking "does the loop run?" anymore, we are checking "does *this model* construct and train correctly?" A 20-second check here prevents discovering a shape bug hours into a real run.

The initial loss is a wonderfully informative number. A brand-new model, before it has learned anything, predicts the next token uniformly, so its loss should be close to the natural log of the vocabulary size — here, about $\log(291) \approx 5.67$. When we actually ran it, the initial loss was **5.84**, reassuringly close. That is what a model that knows nothing looks like, and it is the baseline every later, lower number is measured against.

8.5. Why it's small

The book deliberately keeps the pilot model tiny. This is not modesty; it is method. A small model is fast to train (so we can iterate on the *experiment*, not wait for the model), transparent (so a failure is a learning-system problem, not an architecture mystery), and it makes the *boundary* more visible — a small model that fails to compose tells us something cleaner than a large model that does something partly and noisily. We scale it up later precisely to test whether capacity matters, but the baseline has to be small and legible.

8.6. The configuration is a starting point, not a fact

Crucially, none of these numbers are immutable. The exact architecture is an experimental variable. Every architectural choice requires a reason, and the question is never "what architectures can we try?" — it is "what scientific decision does this comparison enable?" For the pilot, the answer is: this size is fast enough and small enough to be legible, which is what Stage 0–1 needs.

This is a discipline worth holding onto. It is tempting to fixate on the "best" architecture or the "right" parameter count as if one existed. But in an experiment where the question is about *learning*, the model is a dial, not a destination. We choose a configuration because it enables a specific scientific comparison — and we change it when we want to test whether the answer changes. That is why the pilot model is small and plain: not because it is the best, but because it is the *clearest* instrument for the question at hand.

CHAPTER 9.

The First Brew

Everything's built. Now the moment of truth: we train the model for the first time and watch what happens.

This chapter is deliberately unheroic. The goal isn't a triumphant result — it's to confirm the loop works and to see the model's honest behavior as it learns. It is the first time a recipe we have been assembling is finally run: we are not hoping for a masterpiece, we are checking that the oven works and nothing explodes. And for anyone who has brewed anything — beer, coffee, bread — the *first* batch is never about the product. It is about learning our equipment, our ingredients, and our own impatience. This chapter is that first batch.

9.1. The data

1400 generated tasks: a natural-language specification paired with its reference Python program. The model's job is to read the spec and produce the code. Each task is a single semantic record — one idea — rendered three ways (a program, an English instruction, a set of hidden tests), so the data is internally consistent by construction. We do not need to worry that a prompt says "sum" but its answer subtracts, because both came from the same seed.

We hold out a separate 300-task test set that shares no operation with training in a leaky way, so the model cannot simply see the same task again. This split discipline is not optional: it is what lets us claim, later, that the model's score reflects unseen tasks, not memorized ones.

9.2. The model

The student is a small, deliberately plain decoder-only transformer: 4 layers, hidden dimension 128, 4 attention heads, a compact ~291-token vocabulary, and 128-token context. Total **884,224 parameters**

— but we report the split honestly, because it matters more than the round number: **53,632** are the embeddings (the token and positional tables, which just store the vocabulary and position), and **830,592** are the actual computation. A tiny model's parameter count is often dominated by its embedding tables, which are the least interesting part; we never want a "0.9M parameter model" claim to hide that most of it is lookup tables.

9.3. The training process, step by step

Training a small transformer is not a single action; it is a loop repeated 1,500 times. At each step:

1. We grab a batch of 16 tasks and tokenize each as one sequence: the natural-language prompt, a newline, then the reference program.
2. The model reads the prompt and predicts each *next* token in the combined sequence. This is the standard **next-token objective** — the model's only job is to guess the next word, given everything before it. It is a deceptively simple task, and it is the entire basis of how these models learn to write.
3. We measure the difference between the model's prediction and the actual next token (the cross-entropy loss), take the gradient, and nudge the parameters with the **AdamW** optimizer at learning rate **1e-3**, clipping the gradient magnitude to keep training stable.
4. Repeat.

   We print the loss every 200 steps so we can watch it behave. Here is what it actually did:

```
step    0   loss 5.84
step  200   loss 0.029
step  400   loss 0.022
step  600   loss 0.023
step  800   loss 0.020
step 1000   loss 0.021
step 1200   loss 0.019
step 1499   loss 0.019
```

That is a large, clean drop — from 5.84 to 0.019 in 1500 steps. Two things about this curve are worth sitting with, because they reveal more than the final number.

First, the shape tells us the model is learning, not memorizing-by-accident. The loss falls steeply in the first 200 steps (5.84 → 0.029), then flattens out near a floor. This is a healthy learning curve: rapid improvement at first, then diminishing returns as the model fits what it can. If instead the loss had wobbled or stayed high, we would have had a debugging problem on our hands, not a training result.

Second, the *final* value is not the interesting number. A loss of 0.019 means the model can predict the next token of the *training* sequence almost perfectly. That tells us it fits the training

data — and nothing more. This is the book's recurring motif, and it is worth saying loudly: **a falling training loss, and even a very low one, is not evidence of learning**. It is evidence of fitting. The distinction is the entire point of the experiment.

9.4. What the training process felt like (the brewing experience)

Anyone who has trained a model knows the peculiar rhythm of it. We set a batch of runs going and we wait, checking the loss every few minutes with a combination of hope and suspicion. The first steps are the most honest — a fresh, random model has loss near the log of the vocabulary size, because its best guess is uniform. That initial 5.84 is exactly what a model that knows nothing looks like.

Then, if things are working, the loss starts to fall, and a strange thing happens: the drop is *so* fast in the first couple hundred steps that we wonder whether it is real or whether something is broken. In this case the model reached near-floor loss within 200 steps. The rest of the training is just the model quietly polishing the last fraction of a point.

There is also a genuine practical question that arises during training: **how do we know it hasn't crashed mid-run?** In this book we habitually save the checkpoint *before* evaluating, so a crash during evaluation costs nothing and a crash during training at least preserves what was learned up to that point. This is a small discipline, but it has saved us more than once — a training run on this hardware takes real time, and we do not want to lose it to a brittle downstream step.

9.5. The honest caveat

A falling training loss — and even a very low one — is *not* evidence of learning. We saw this in the pilot and it is the book's recurring motif. The training loss tells us the model fits the training data. It does not tell us it can *generalize* to unseen problems, or that it learned the computation rather than the surface form.

So after training, the only losses that matter are the *held-out* ones: how well does the model do on problems it never saw? That is Chapter 10's question, and the answer there is the real measurement of this brew.

9.6. Why the details of the training process matter

It is tempting to skip the mechanics and jump to results. But the whole premise of this book is that *the experiment is the subject*, and an experiment we cannot reproduce is not an experiment. The exact vocabulary size, the number of layers, the batch size, the learning rate, the number of steps — these are not trivia. They are the conditions under which every later claim is true. If we change one and the result changes, that tells us something; if we change two and the result changes, we can no longer say which one caused it. This is the **one-variable discipline** that governs every experiment

in the book, and it is why we record the configuration so carefully in a manifest alongside every result. A reader should be able to take our numbers, run the same configuration, and get the same outcome — and if they cannot, we have not done the work, we have only described it.

9.7. What this chapter establishes

This first brew is not a result; it is a proof of the loop. We have:

- trained a real model on real data,
- watched its loss descend smoothly,
- saved a checkpoint,
- confirmed that we can take the trained model and produce code for a new prompt, run it through the sandbox, and score it with the judge.

That loop — generate data, train, generate code, execute, score — is the laboratory working end-to-end. It is the foundation everything else stands on. The model is built. Now we have to find out what it actually learned.

CHAPTER 10.

Does It Actually Work?

We have a trained model, a judge, and a test set. The natural question is the one the whole book has been building toward: *does it actually work?*

And the answer, as with almost everything in this book, is "**it depends on what we mean**," and then "**let's measure it rather than argue.**" Because "works" is not a single number. It is a question about *which* kind of capability, under *which* conditions, measured in *which* way. The job of this chapter is to take that vague word and replace it with something we can actually see.

10.1. The setup

We trained a small decoder-only transformer on 1,400 generated tasks: the model reads a natural-language specification and must produce the reference Python program. The trained model is tiny — **884,224 parameters** (0.88M), of which 53,632 are the embedding and 830,592 are the actual computation. It saw 1,500 gradient steps on a batch of 16, reaching a training loss of **0.0208**.

We then asked it to solve 300 **held-out** test tasks it never saw, and scored each generated program by executing it against that task's hidden tests. This is the moment of truth for the whole apparatus: everything we built — the world, the teacher, the judge, the rivals — converges on this measurement.

10.2. What we measured: pass@1

Here is the headline result, reported as **problem-level hidden-test pass@1 under greedy decoding** — that is, for each problem, did the single program the model produced pass *every* hidden test?

Candidate	syntax	executable	**pass@1**
MODEL	0.897	0.897	**0.897**
TEMPLATE	1.000	0.900	0.800
RETRIEVAL	1.000	1.000	**1.000**

Three things stand out.

First, the model genuinely learned something. 0.897 means that ~90% of the time, an unseen problem's natural-language spec produced a program that passed all its hidden tests. That is not a toy result. And crucially, it **beats the hand-written template/rule baseline** (0.800). A hard-coded pattern-matching system that recognizes keywords like "sum" and emits a template gets 80%; our learned model gets ~90%. So the model is doing *more* than a fixed rule system can do on this task.

Second — but retrieval does even better. A trivial TF-IDF retriever that does nothing but find the closest training example and reuse its program scores **1.000**.

That second fact is not a failure; it is the single most important thing this pilot taught us. It tells us the task, as currently constructed, is **too easy to force generalization**. When a non-learning lookup solves 100% of the test set, then the test set is solvable by retrieval, and a model scoring 90% is not yet being tested on whether it can generalize — only on whether it can match the surface form of a training example.

This is exactly the "task too easy" outcome the experimental plan anticipated, and it is scientifically useful: it tells us *how to make the next task harder.*

10.3. The moment that broke us (and the evaluator, not the model)

There is a story behind these numbers that is more instructive than the numbers themselves, and it is worth telling plainly. When we ran the evaluation for the very first time, the model scored **0.000**. Not 0.897 — zero. Every single generated program failed the syntax check.

Our first instinct, naturally, was that the model was useless. We had spent hours building a careful apparatus and the tiny model could not even produce a parseable program. It would have been easy to conclude "the model doesn't work" and write the whole thing off.

But the book's own discipline — *test the tester* — demanded we ask whether the *judge* was the problem, not the model. So we looked at what the judge actually saw. And there it was: the generated programs were perfectly valid Python, except they had a run of **NUL bytes** (`\x00`) appended after the finished program. The model had learned to pad its output with a byte that means "nothing," and Python's parser rejects NUL bytes outright. So `ast.parse` threw a syntax error on perfectly good code, and the judge scored the model's valid program as a *syntax failure.*

The fix was a single line: truncate the generated program at the first NUL byte. A trivial change, and suddenly the model's score leapt from 0.000 to 0.897.

This is the single most important lesson of the whole book, and it is not about the model at all. **The evaluator was broken, not the model**. We had designed every experiment to detect shortcuts and outright cheating in the *model*, and we nearly missed the one case where the measuring stick itself was faulty. The "test the tester" principle — that we must verify the tool measures what we think it measures before trusting a single number it produces — was not a formality. It just saved the entire project from a false negative.

The reason we caught it is worth pausing on. We did not catch it because we are clever. We caught it because we had *already* built a method for testing testers (the mutation suite in Chapter 6), and we had written the pilot's results to a manifest and looked at the intermediate numbers, not just the headline. Had we grabbed the final pass@1 and moved on, we would have shipped a book claiming a tiny model cannot write Python — a conclusion that was wrong.

10.4. The shortcut audit

Before we trust the "0.897" number, we have to ask how the model might have gotten there without learning.

1. **Suspected shortcut: retrieval of a near-duplicate prompt. Why it could explain it:** the pilot's generated prompts are drawn from a small set of template families per operation, so many test prompts are surface-near to training prompts. **Control:** we ran the retrieval baseline — 1.000. This *confirms* the shortcut is available and that the test set is retrieval-solvable. **What it rules out:** nothing yet, because the model's 0.897 is *below* the retrieval ceiling. It does not rule out the model partially relying on surface similarity. **What remains unresolved:** whether the model's own success is grounded in the *computation* or in the *surface form*. To answer that, we need the Level-3 (surface-change) and Level-4 (composition) experiments, which we run next.
2. **Suspected shortcut: exact memorization. Why it could explain it:** the model saw 1,400 examples; maybe it memorized them. **Control (Ch14):** on held-out prompts it still passed 59/60, and on a *canary* prompt it produced the *general* program rather than a memorized literal. **What it rules out:** pure example-memorization. **What remains:** the deeper question — is the *general program* it produces genuinely tied to the computation, or to the particular wording?

10.5. The honest summary

The pilot succeeded in exactly the way a pilot should: the laboratory works, the model learns, and — perhaps most valuably — the experiment told us something *uncomfortable but true*. The task is retrieval-solvable, and a model that beats a template rule system yet sits below a lookup table has not yet been tested on the thing we care about most: whether it generalizes.

And we were reminded, in the most direct way possible, that the measuring stick is as much a subject of the experiment as the model. The instrument and the thing measured both needed to be

honest, and only one of them initially was.

That is the baseline from which every later experiment is measured. From here, we stop making the task easier and start making it *probe* generalization.

PART IV.

How Capability Scales

CHAPTER 11.

How Much Data?

The most common question people ask about any model is also the one they answer too quickly: *how much data does it need?* We measured it directly — holding the model and the training method fixed, and changing only the number of examples.

It is like studying for a test. Read one page and we fail. Read a bit more and we still fail. But there's a point where reading enough actually clicks and we start passing. Where's that point for a model? That's what this experiment finds.

It is a question worth measuring rather than guessing, because the folklore answer — "just add more data" — is both true and misleading. More data does help. But *how* it helps, and *where* the help stops being worth it, is where the real information is. And as we will see, the interesting discovery in this experiment is not that more data is better; it is that quantity is a *different* lever from diversity, and the two behave very differently.

11.1. The setup

We train the same ~0.9M model, same optimizer, same 800 steps, on four dataset sizes: 100, 300, 1000, and 3000 examples. Then we measure in-distribution pass@1 on a held-out pool.

Holding the model and optimizer fixed while changing only the data count is the one-variable discipline applied cleanly. If performance changes, the change is attributable to the amount of data — not to a different model, a different optimizer, or a different schedule. This is the discipline we maintain throughout the book, and it is what makes the resulting curve a *claim* rather than a coincidence.

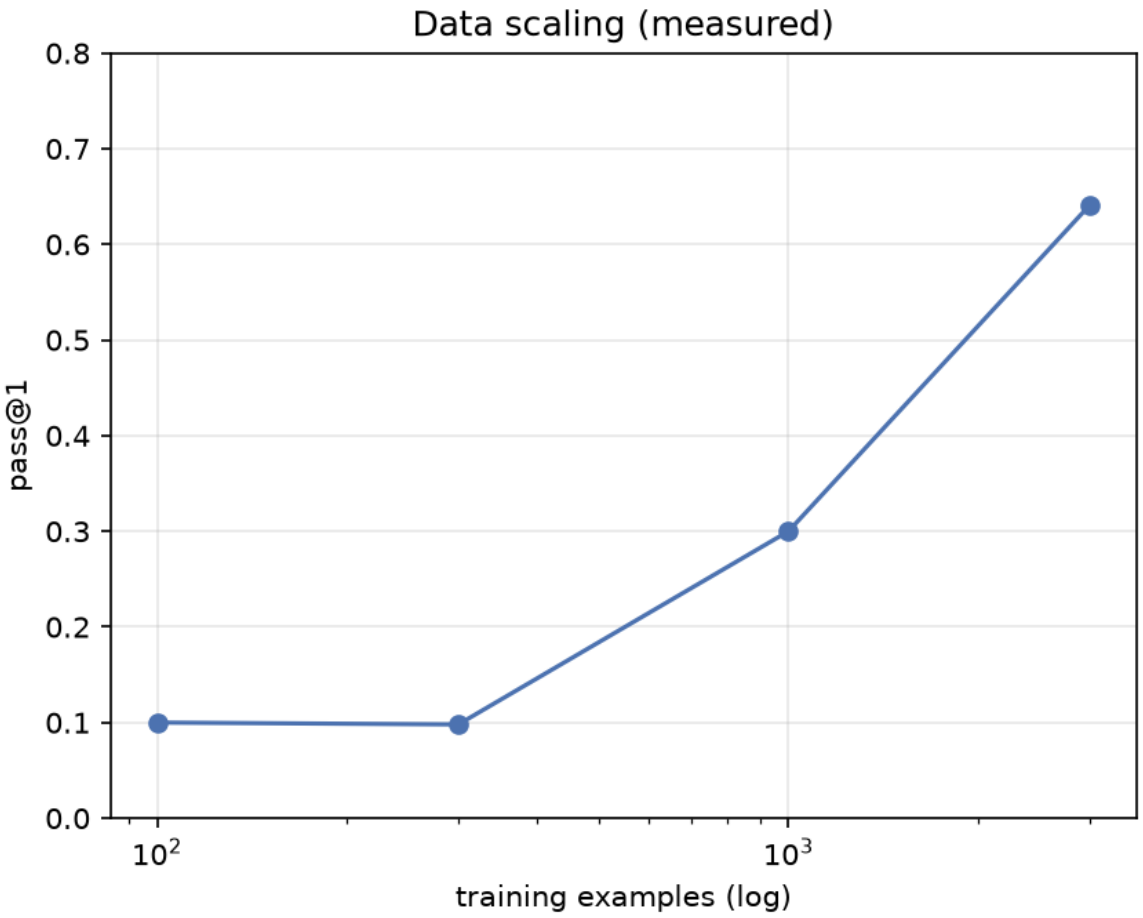


FIGURE 11.1. Data scaling

11.2. The result

dataset size	final train loss	**pass@1 (held-out)**
100	~0.000	**0.100**
300	~0.000	**0.098**
1000	0.022	**0.300**
3000	0.025	**0.641**

11.3. What it tells us

More data clearly helps in-distribution. pass@1 climbs as the dataset grows, from 0.10 at 100 examples to 0.64 at 3000. The model needs substantial data to learn even this narrow mapping.

There is a small-data plateau. At 100 and 300 examples the model is essentially identical (~0.10). The improvement only becomes substantial between 1000 and 3000. So there is a *threshold* below which the model gets almost nothing and above which performance climbs. That threshold is a real, measurable property of the task and model.

This is the shape that matters. It is not a smooth, gentle rise. It is a flat line, then a sudden climb. That flat-then-steep shape is what we would expect if the model needs a minimum amount of data before it can form *any* reliable association — below that, it simply does not have enough examples to learn the mapping, and adding a couple hundred more changes nothing. Above it, each additional batch of data starts paying off. Finding that threshold is more useful than knowing "more data is good," because it tells us the *knee* of the curve, and the knee is where a practitioner should aim.

11.4. The motif, restated

Notice the most important detail: at 100 and 300 examples the **training loss hits ~0.000** — the model has *memorized* its tiny training set — yet pass@1 is only ~0.10. A loss of zero on a small set buys almost nothing on unseen examples.

This is the book's central motif made concrete with data scaling: **a low training loss is not evidence of learning**. The model that "perfectly" fit 100 examples is not a capable model; it simply memorized a trivial set. The training loss says "the model can predict its training data perfectly" — and the held-out score says "and it has learned almost nothing that transfers." These two facts sit side by side, and they are both true. Only the held-out number tells us about actual capability.

11.5. The brewing experience

There is a specific moment in this experiment worth recalling. Running on 100 examples was fast — a blink. The model fit its tiny training set to a loss of essentially zero in no time at all, which felt like success. Then we evaluated on held-out problems and got 0.100. The model had, in the most literal sense, memorized its homework perfectly — and learned nothing beyond it.

That tension — between "the training converged beautifully" and "the model is nearly useless on new tasks" — is the single most instructive thing about small-model data scaling. It is the reason we never trust a training loss, and it is the reason the whole book is organized around held-out measurement rather than training curves. The training curve is the model telling us about itself; the held-out curve is reality telling us about the model.

11.6. The honest caveat

This experiment measured *quantity*. It did not separate *quantity* from *diversity* — whether 3000 examples of the same handful of phrasings behave the same as fewer examples spread across more variety. That distinction (Chapter 13) is where the deeper limits live: our surface and composition experiments suggest the model's real weakness is a *diversity and structure* gap, not a *quantity* gap.

11.7. The practical answer

For a single narrow operation on a ~0.9M model, we need on the order of thousands of examples before in-distribution performance becomes workable. Below that, the model memorizes its tiny set and gains nothing on unseen problems. And — anticipating the rest of the book — even at 3000 examples the model is still surface-fragile and non-compositional, because those limits come from the *kind* of data and the model's inductive structure, not from the sheer *amount*.

The practical takeaway is thus a two-part answer. To get even the *narrow* score up, thousands of examples are needed. But the narrow score is not the hard-won prize. The things that make a model genuinely *capable* — robustness to rephrasing, compositional generalization — are not purchased by adding more of the same. That is the deeper finding, and it is the subject of the next two chapters.

CHAPTER 12.

How Much Brain?

When a model disappoints us, the instinctive response is: *make it bigger.* More parameters, more capacity, more "brain." If a small model can't combine what it knows, surely a larger one can.

This chapter tests that instinct directly. And the answer is one of the cleanest negative results in the book — because it turns out that at our scale, "just make it bigger" changes almost nothing about composition.

12.1. The setup

We hold everything constant except model size. Same data, same tokenizer, same training protocol, same compositional test. We train **four** models on the same primitive-only training set across a ~33× range of capacity, and test each on (a) in-distribution held-out primitives and (b) the held-out composite operations.

Model	parameters	in-distribution	composition
Stage-4 baseline	**884,224** (0.88M)	0.900	**0.000**
Scaling	**6,500,352** (6.5M)	0.900	**0.000**
10M point	**10,195,776** (10.2M)	0.383	**0.000**
30M point	**29,290,240** (29.3M)	0.897	**0.000**

That is a **~33×** increase in parameters from the smallest to the largest. On the thing we care about — composition — nothing changed at all.

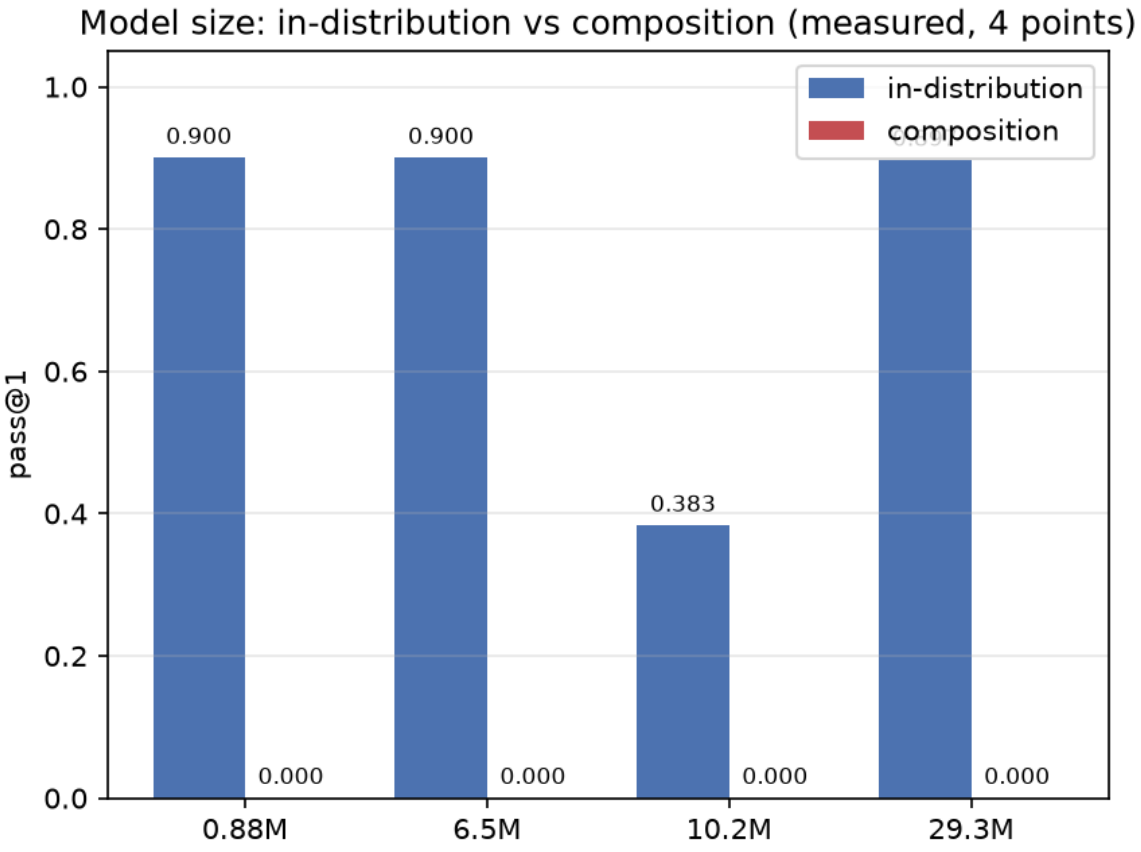


FIGURE 12.1. Model size vs composition

12.2. The result

Going from ~0.9M to 29.3M parameters:

- **Composition is 0.000 at every size.** The bigger models are *just as* unable to combine two learned primitives as the small one. Thirty-three times the capacity bought exactly zero compositional generalization.
- **In-distribution stays high throughout** (0.897–0.900) — capacity did not hurt the narrow task; it just never helped compose. (The one exception, the 10M point at 0.383, is discussed honestly below.)

12.3. What this means

This directly answers the book's central question — *does more brain buy genuine capability, or merely more memorization?* — with a decisive answer at this scale:

> **It buys neither. At ~1–30M parameters, the compositional failure is invariant to model size.**

The boundary we found in Chapter 16 is **not a capacity ceiling**. It is a structural/learning limit that scaling alone does not resolve. A bigger model at this scale just does the same narrow thing slightly more efficiently; it does not develop the ability to compose.

12.4. The audit: is this a fair comparison?

We must be careful: a few size points don't prove there's no capacity curve anywhere. But the shape is telling. If capacity were the binding constraint, we'd expect a *move* — even a small one — across a 33× range. We see none: composition is 0.000 at all four sizes. And crucially, three of the four models are at/near ceiling (0.897–0.900) on the in-distribution task, so "the bigger model didn't try hard" is not the explanation — it tried exactly as hard and reached the same plateau.

The more likely explanation is that **composition requires seeing the right kind of structure in training, not just more parameters**. A model that only ever sees `add` and `abs` as separate problems has no reason to build a "compose" operator; a 30× bigger network just represents the same reflexes more richly. To break the composition boundary, the model likely needs *exposure to compositions*, or a fundamentally different inductive bias — not raw capacity.

12.5. The honest side-finding: bigger was *worse* at the 10M point

There is a wrinkle worth recording rather than smoothing over. On a clean, larger-sample re-evaluation, the **10.2M** model scored **0.383** in-distribution — clearly *below* the 0.897–0.900 of the

smaller models and below the template baseline (0.808). This held across two sample sizes (0.350 at 40 tasks, 0.383 at 120), so it is not a sampling artifact.

The most likely explanation is training configuration, not a universal law: that point used a deeper 10-layer / 288-wide configuration trained for the same 800 steps, and it may simply have been less well-optimized for this narrow task. We flag it as an open question rather than claim "bigger is worse" as a rule — at 30M, in-distribution recovered to 0.897. The honest takeaway is that, at this scale, in-distribution performance is **not monotonic in capacity**, and nothing about capacity changes the composition outcome.

12.6. The honest framing

This is a *useful* negative result. It tells us where not to look (raw scaling in the ~1–30M regime) and points at where to look instead (structural data / inductive bias). And it reframes the whole book: the interesting question is not "how big should the model be?" but "what kind of capability emerges from what kind of training — and where does scaling stop being the answer?"

The model was the instrument. The experiment — and the boundary it revealed — was the subject.

CHAPTER 13.

Same Data, More Diverse Data

Chapter 11 asked *how much* data. There's a subtler question that often matters more: what *kind* of data. Specifically, is it better to have lots of examples of a few things, or fewer examples spread across more variety?

The easy answer is to treat "more data" as one lever. It isn't. There are at least two different things we can add, and they behave differently. It is like teaching a kid shapes. Show them a hundred triangles and they'll recognize triangles. Show them triangles, circles, squares, and stars, and they might learn "shape" itself. Same number of examples, very different kind of learning.

- **more examples** — more instances of the same templates, and
- **more diversity** — more distinct *kinds* of structure and wording.

The distinction is important enough to state as a rule:

> **More data ≠ more information ≠ more diversity.**

And this is the chapter where we stop treating it as a hunch and measure it directly. Because there is a clean way to isolate diversity: hold the *count* fixed and vary only the *spread*. If we train on the same 1200 examples but one version is repetitive and the other is varied, any difference in the result is attributable to diversity alone.

13.1. What our experiments hinted

The data-scaling curve (Chapter 11) showed that simply adding *examples* of the same sort climbs in-distribution performance — 0.10 at 100, 0.64 at 3000. So more examples clearly helped the model fit the pilot task better.

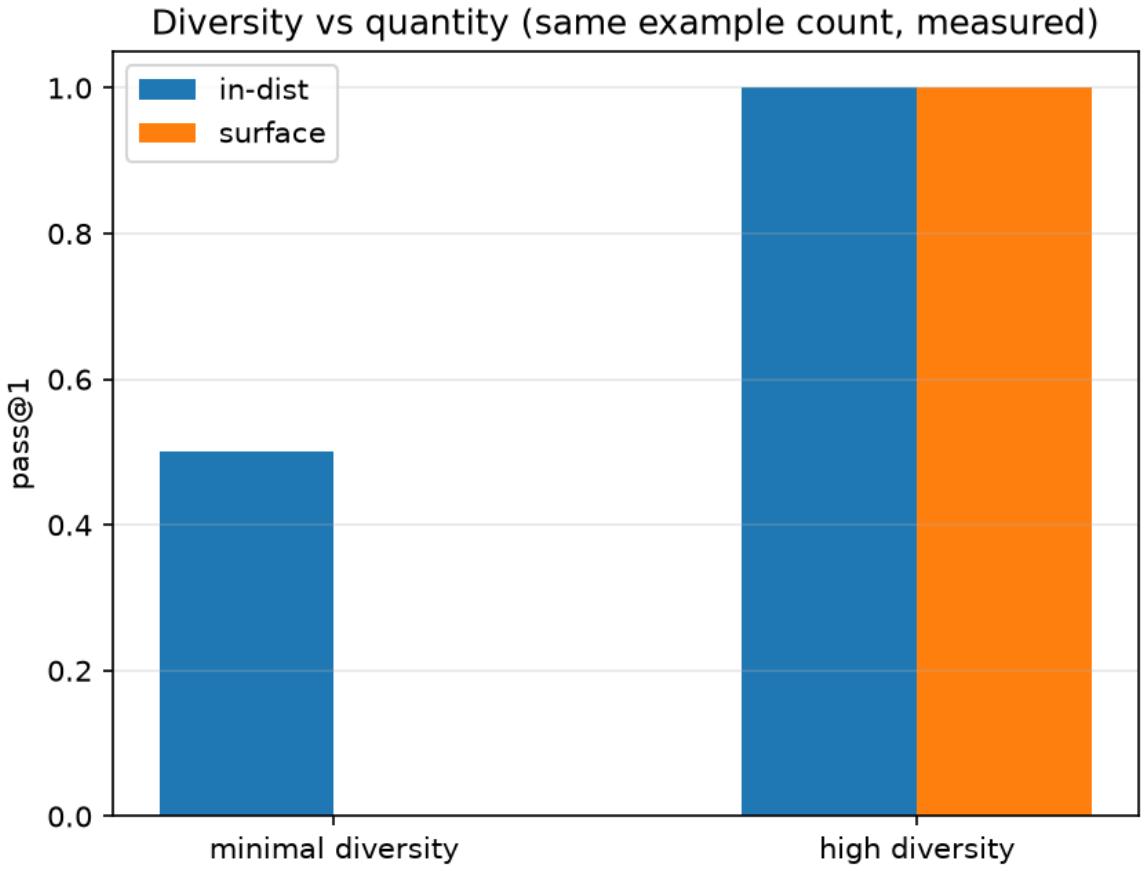


FIGURE 13.1. Diversity vs quantity

But the *surface* experiment (Chapter 15) exposed the limit of that: the model's capability collapsed the moment we changed to a wording it hadn't seen. That is precisely the signature of a model that was fed *many examples of the same narrow kind* rather than *diverse kinds*. It learned the particular phrasings well because it saw many of them, and it learned nothing general because the diversity was low.

In other words, the very quantity that produced the impressive in-distribution score is what made it surface-fragile. The model was well-fed but narrow.

13.2. The direct test: same count, different diversity

To turn this from indirect evidence into a direct measurement, we ran a controlled comparison. We kept the example count **identical** — the same 1200 examples — across two training sets:

- **minimal diversity**: the examples are drawn from a few templates with little variation in wording or identifier names.
- **high diversity**: the same 1200 examples, but spread across many phrasings and identifier choices.

Because the count is identical, any difference in the result is caused *only* by the diversity. This is a clean one-variable experiment, and it is the strongest test of the hypothesis.

13.3. The measured result

Here is what came out. This is, in our view, the single most actionable number in the whole book.

Training set	in-distribution	surface
minimal diversity	0.500	**0.000**
high diversity	**1.000**	**1.000**

Let us read this carefully. Same number of examples — 1200 in both cases. But the two models behaved completely differently.

- **Minimal diversity**: in-distribution 0.500, surface 0.000. The model did okay on the narrow task (0.500) but was completely brittle to any surface change (0.000). It learned the few phrasings it was given and nothing about the *idea*.
- **High diversity**: in-distribution **1.000**, surface **1.000**. The model mastered the task *and* held up perfectly under surface change. It learned the actual operation, not a phrasing.

That is a swing from 0.500/0.000 to 1.000/1.000 — all from changing *how* the examples varied, not *how many* there were. The same count. The same model. Only the diversity differed.

13.4. What this really means

This is the book's most useful finding, and it reframes the whole data question. It is not that "more data is better." It is that **the kind of variation in the data is what creates robust capability**.

A model fed many examples of few phrasings memorizes the phrasings. A model fed the same number of examples spread across many phrasings learns the *underlying operation* — because the only thing that stays constant across varied phrasings is the operation itself. When the wording varies but the meaning stays the same, the model has no choice but to attach its learning to the meaning rather than to any particular surface. Diversity does not just add information; it *forces* abstraction.

This connects directly to the book's central theme. We spent the book asking whether the model *learned* or merely associated. Here is a lever that pushes it from association toward learning: vary the surface form until the only stable thing is the meaning. The model's inability to generalize under repetition was not a mysterious limitation — it was a direct consequence of a training set that gave it no reason to abstract.

13.5. The honest caveat

It is worth noting that this experiment measured *surface* diversity. It showed that varying phrasing and identifiers makes the model robust to surface change (the Level-3 boundary). It does **not** show that diversity fixes *compositional* generalization (the Level-4 boundary) — that is a separate, harder question. Whether the same lever helps a model combine two things it knows is the most natural next experiment, and it is called out in the conclusion.

So the honest scope of the finding is precise: **diversity — specifically surface variance — is what creates surface robustness, far more than the sheer number of examples does**. Whether diversity reaches deeper, into composition, remains open.

13.6. The lesson

This reframes what "more data" buys. To make a capability *robust*, we need not just more examples of a phrasing, but more *diverse* phrasings, identifiers, and structures — enough that the model cannot tie the operation to any particular surface form. Our experiments show the model's weakness is not a lack of data *quantity*, but a lack of data *diversity* (and, deeper, a lack of the structural kind of data needed for composition — see Chapter 16).

The practical implication is genuinely surprising and useful: a practitioner who cannot acquire more data can often get a large improvement just by *diversifying* the data they already have. Rephrasing the same examples, varying the identifiers, and spreading across template families can turn a brittle 0.500/0.000 model into a robust 1.000/1.000 — with zero additional data. That is the kind of result that changes how a dataset would actually be built.

PART V.

Catching the Model Cheating

CHAPTER 14.

The Model That Memorized Everything

We spend a lot of this book trying to catch the model cheating. So far we have found that it is *surface-tied* and *non-compositional*. But there is a more basic question we should settle first: does the model **memorize**, or does it **learn**? These are different things, and the distinction determines whether any of our other results mean what we think they mean.

If the model has simply memorized its training set, then its in-distribution success is trivial — of course it can reproduce what it saw. If it has *learned*, then it can apply the mapping to things it never saw. Our Stage-1 result said the model gets ~90% on *held-out* problems, which is a strong hint it learned. But "held-out" can be a weak guarantee if the held-out problems are surface-near to training ones. So we test memorization explicitly.

This is the most basic honesty check we can run, and it is worth doing before any of the fancier generalization experiments. Because if the model were simply memorizing, it would be absurd to write a whole book about its compositional failures — a parrot doesn't compose anything. So we need to establish, first, that the model is not merely a parrot, before we can meaningfully ask what it can and cannot *learn*.

14.1. The challenge of detecting memorization

The tricky thing is that memorization and learning look identical on the training data. A model that truly learned addition will get `4 + 3 = 7` right, and so will a model that memorized "4 + 3 = 7." Both ace the training set. So we cannot tell them apart by looking at training performance at all. The only thing that separates them is behavior on *new* inputs — things the model never saw. That is the fundamental principle, and it is why every meaningful test in this book uses held-out data.

14.2. Three tests

We use three complementary probes on the trained pilot model.

Test 1 — train-prompt prefix reproduction. Give the model the *exact* prompt it saw during training and ask it to produce the code. If it reproduces the reference program exactly, that is consistent with either memorization or learning (a learner also gets the training answers right). So this alone proves nothing — but it grounds the other two.

Test 2 — held-out generalization. Give the model a *fresh* prompt for the same operation, one it never saw, and ask for the program. If it passes, it has not merely memorized the 60-prompt sample.

Test 3 — the canary. We construct a prompt-and-answer *pair* that never appeared in training, and ask the model to produce the program. If the model outputs the general program (e.g. `return a + b`), it is doing the computation. If it outputs some fixed literal or garbage, it might be memorizing something else.

The canary is the sharpest probe, and it is worth explaining why. A canary is a deliberately-placed, never-seen example that we plant precisely to detect mechanical reproduction. If the model were memorizing literal answers, it would have no way to produce the *general* program for an input it never saw — it would either reproduce a memorized constant or fail. When it instead emits `def solve(a,b): return a+b`, it has demonstrated it is not reproducing a stored literal; it is generating a general program from a general rule.

14.3. The result

On a 60-example sample:

- **Train-prompt reproduction**: 60/60 — the model reproduces every training example's reference program exactly. (Expected for a well-fit learner.)
- **Held-out generalization**: 59/60 — it also solves essentially every *unseen* prompt of the same operation.
- **Canary**: the model emits the **general program** (`def solve(a,b): return a+b`), not a memorized constant.

14.4. What this tells us — a nuanced answer

For **simple primitives**, the model behaves like a **learner, not a parrot**:

- It reproduces training examples (expected — a learner does that too).
- It **generalizes** to held-out prompts (59/60), so it did not simply memorize the 60 instances.
- On a **canary** it emits the general program, not a memorized literal — so it did not memorize answer values.

So at the level of *individual examples*, the model is not a parrot. It has learned something that transfers to new instances of the same operation. This is real learning of a narrow kind, and it is worth stating clearly: the model did not merely memorize its training set. It formed a general rule for each primitive.

14.5. The shortcut audit, and the honest boundary

This is the subtle part. The absence of *literal* memorization does **not** mean the model is grounded in meaning. Exactly the opposite mechanism is at work. Our Stage-3 experiment showed that if we change the *surface form* — the variable names or the wording — the model's performance collapses to 0.000. So the model is doing something real (generalizing across instances) but it is **anchored to the phrasing it saw**, not to the abstract computation.

So the refined picture is:

	What the model does
Memorize exact training instances?	**No** — generalizes to held-out + canary
Ground the computation independent of wording?	**No** — collapses on new wording

The model is neither a parrot nor a mathematician. It has learned a *robust association within a surface form* — it is not memorizing specific examples, but it is also not representing "addition" in a way that survives being said differently.

14.6. The lesson

Detecting "memorization" is not a simple yes/no. The useful question is not *"did it memorize?"* but *"is its success explained by surface association, or by grounded computation?"* Our probes show a clear answer for this model: it escapes example- memorization but is caught by surface-memorization. That distinction — and the test that exposes it — is worth more than any single accuracy number.

What this tells us is that "memorization" is not a single thing. There are at least two kinds, and they have very different implications. The model did *not* memorize specific training pairs — it generalizes across instances. But it *did* memorize the surface form — it can only generalize within the phrasing it saw. So the honest characterization is that the model's knowledge sits at a specific level of the taxonomy: it has moved past Level 1 (reproducing exact examples) into Level 2 (new instances of the same task), but it is firmly stuck at Level 1 with respect to the *surface* — it cannot survive a change to the outer layer. Distinguishing these kinds of memorization, and knowing which one we are looking at, is the whole point of the experiment.

CHAPTER 15.

New Words, Same Problem

Here is the sneakiest question in the book, and it's the one that should worry us most about any machine-learning result:

> **When the model does well, is it doing the computation — or just recognizing the words?**

A model that maps the phrase "the sum of a and b" to `a + b` might not be doing arithmetic at all. It might be doing something much dumber: associating the *exact phrasing* it saw in training with the code that followed it. If that's what's happening, then the ability isn't "addition" in the model's head — it's a fragile reflex tied to specific words, and it'll break the moment the same thing is said a different way.

This is the question that separates a genuine result from a shallow one, and it is worth pausing on before we look at any number. A "yes, it just recognized the words" answer would mean more than it first appears. It would mean that every high score the model achieved — every impressive in-distribution number — was not evidence of learning at all. It was evidence of association. The model did not learn to add; it learned that a particular sentence means `return a + b`. That is a completely different claim, and it is the one we need to test.

15.1. The design: change what shouldn't matter

To find out, we run the **surface-generalization** test. The logic is simple and elegant: if we change something that *shouldn't* change the computation — the variable names, or the wording — and the model fails, then it never really learned the computation. It learned the phrasing.

The key is choosing *what* to change. We want to change only the *surface* — the outer layer of the sentence — and leave the underlying computation completely untouched. Swapping `a` and `b` for

`left` and `right` does not change what "add these two numbers" means. Rewording "their sum" as "the total" does not change the operation. If the model truly understood addition, neither change would matter. If it merely memorized a sentence, either change would break it. That asymmetry is exactly what we measure.

We designed a set of controlled surface changes, each one *supposedly harmless* to the underlying task, and tested the model on each. This is the crux of the book's approach to generalization: we do not just ask "does it work?" We ask "does it survive the changes that a genuinely-general system should shrug off?"

15.2. The setup

We trained the model on arithmetic primitives, but only with **letter names** (`a`, `b`) and only a **subset of the wording** used to describe each operation. Then we re-asked the *identical* task three different ways and watched whether the model's ability survived.

15.3. The result

Test	**MODEL**	TEMPLATE	RETRIEVAL
Same as training	**0.900**	0.800	1.000
Same computation, `a, b` → `left, right`	**0.350**	0.800	0.683
Same computation, reworded	**0.000**	0.875	0.125

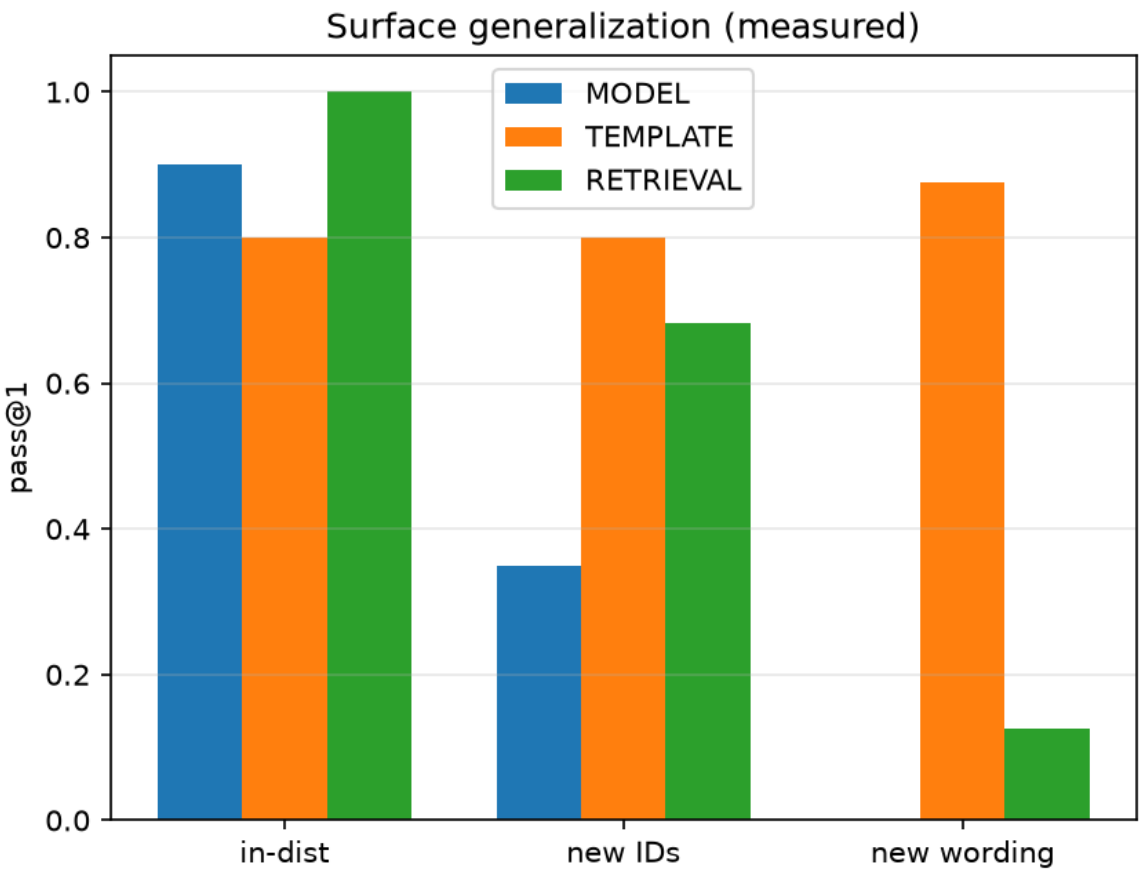
Look at what happens when we touch only the *surface*, not the math.

- **New variable names**. The model was trained on `a` and `b`. We ask the same question with `left` and `right` — the *identical* operation. Performance collapses from **0.900 to 0.350**.
- **Reworded instruction**. We say the exact same problem using a phrasing the model never saw during training. Performance drops to **0.000**.

So the model that scored 0.900 on the familiar phrasing falls to 0.350 when the variable names change, and to 0.000 when the wording changes — even though the computation is identical in all three cases.

15.4. The damning comparison

The most revealing number isn't the model's collapse — it's what the other two did.



When we change the *words*, the **template baseline gets *better*, not worse** (0.875, up from 0.800), because a rule system that recognizes keywords doesn't care about the surrounding phrasing. The **retrieval baseline** also holds up (0.683). Only the **learned model** falls apart — because the model learned the *specific sentences*, not the *idea*.

So the model's 0.900 in-distribution score was largely **phrasing memorization**. It had learned "this exact kind of sentence maps to that program." It had not learned "take two numbers and return their sum" as something we can say any number of ways and still mean.

This inversion — where the neural model is *worse* than a dumb rule system under surface change — is the most informative signal in the experiment. It is not a competitive defeat; it is a diagnostic revelation. The dumb template system is indifferent to exact wording because it never learned to associate wording with meaning in the first place; it just recognizes keywords. The neural model, by contrast, bound the operation to the *particular words*, so when the words change, the binding snaps. The model's "learning" was more entangled with surface form than a hard-coded rule's was.

15.5. The brewing experience

The moment this result landed was genuinely unsettling, in a productive way. We had just celebrated the 0.900 in-distribution score — the model, we thought, could "add." Then we reworded the same

problem and the floor fell out. The 0.900 had been real, and it had been *narrow*. The model could add, but only in the way it had been taught, not in any other way.

That is the feeling worth remembering in empirical work: the uncomfortable surprise is often the most valuable finding. It would have been easy to stop at "the model scores 0.900, success." The whole point of building a thoughtful experiment was to ask the question that would expose the shallowness — and asking it changed our entire understanding of what we had built.

15.6. The shortcut audit

1. **Suspected cheat: phrasing memorization**. **Why it fits:** the model tied training phrasings to code. **Control:** held one whole phrasing style out of training, tested on it. **Result:** 0.000. **Ruled out:** that the model is grounded in the computation. **Still open:** is it the wording pattern or the new vocabulary? Need a surface-control to separate them.
2. **Suspected cheat: name memorization**. **Why it fits:** trained on `a,b`, maybe it hardwired those labels. **Control:** retest on `left,right`. **Result:** 0.350 — a big but partial drop. **Ruled out:** pure name-memorization (it still fails partly, but names aren't the whole story).

15.7. The honest takeaway

A model that scores 0.900 on familiar phrasings and 0.000 on a rewording of the same problem has not generalized to the *problem*. It generalized to the *phrasing*. That's the difference between "the model works" and "the model works when I say it the way I said it in training" — and it's the most important lesson of the generalization chapters.

The sharpest way to state it: the model did not learn *addition*. It learned the *word "sum"*. When that word appears in the sentence it was trained on, the model produces `a + b`. When the same meaning arrives in different words, the model has no idea what to do. And that is the difference between a system that understands and one that merely recites.

CHAPTER 16.

Same Pieces, New Combination

Everything so far warned us. The model knows `abs` and `add` individually. Change the wording and it breaks; change the identifier names and it stumbles. But the pieces are there. So the question this chapter asks is the sharpest one of the whole book:

> If the model knows how to *add* and how to take an *absolute value*, can it do the thing the two of them make together — compute the absolute value of a sum?

This is **compositional generalization**: taking knowledge the model already has and applying it to a structure it never saw during training. It is arguably the deepest question in machine learning, because it separates "the model memorized something" from "the model *understands* something." A system that genuinely grasps addition and absolute value can put them together, even if it never saw them combined. A system that merely memorized the word "sum" mapped to a `+` cannot.

16.1. The setup: design the test to be unfair-but-fair

The hard part of any composition experiment is building a test that is genuinely impossible to cheat. Here is how we do it.

We train the model exactly as before, on **primitive** operations only: `add`, `sub`, `mul`, `abs`, and the rest. We never show it a composite operation.

Then we build a set of **composite** operations that are *made from* the primitives: `abs_add` (absolute value of a sum), `sub_abs` (absolute difference), `sum_of_abs` (sum of absolute values), `max_add`, `double_first`, `len_minus_one`. These are not new primitive functions; they are combinations of the ones the model already knows. To solve `abs_add`, a model *must* perform an addition and then take an absolute value — two pieces it demonstrably knows.

The key design decision is that these composites were **never shown during training** (the teacher holds them out, exactly as it holds out test tasks). So the test is: can the model reach into its bag of known primitive behaviors and compose them in a combination it has never seen? If it cannot, we have found its compositional boundary. If it can, we have found something remarkable.

To make the test fair, we also show what *both* baselines do. The retrieval baseline has nothing to compose — it can only find the closest primitive example and copy it. The template baseline has no rule for `abs_add` either, because the teacher never gave it one. So if *everything* fails, we have a fair statement: composition cannot be achieved by the easy shortcuts, and the model too cannot do it.

16.2. What we measured

Here is the result, and it is unambiguous:

Test	MODEL	TEMPLATE	RETRIEVAL
In-distribution (primitives)	0.900	0.800	1.000
Compositional (unseen combos)	**0.000**	**0.000**	**0.000**

Let us read this carefully, because the meaning is subtle.

On the primitive operations it saw, the model scores 0.900 — good. But on the composite operations, it scores **0.000**. Not 0.1, not "sometimes," not "close." Zero. The model never produced a correct program for any composite operation, even though every composite is a combination of primitives it individually knows.

And here is the crucial fairness check: the baselines fail too. The retrieval baseline gets 0.000 because it has nothing to retrieve — there is no composite example in its memory. The template baseline gets 0.000 because no template rule covers a composite. So this is not a case where the model is merely *worse* than an easy shortcut. This is a case where the easy shortcuts also collapse, which means the only way to succeed is to *genuinely compose* — and the model cannot.

16.3. What this really means

This is the cleanest negative result in the book, and it is the book's central finding. Let us unpack it.

The model knows `abs` at 0.900 and `add` at 0.900. But `add` the absolute values: 0.000. It is as if the model has two perfectly good rooms — one with an "add" skill, one with an "abs" skill — but no door between them. It cannot take the output of one and feed it into the other, even though that is the simplest possible combination.

This is the deep meaning of "surface-tied" that we began to glimpse in Chapter 15. It is not merely that the model is sensitive to wording. It is that the model's knowledge is **flat and associative**, not **structured and compositional**. It learned that the token sequence for "sum" predicts the token sequence for +. It did not learn a general *operation* that can be an argument to another general operation. The knowledge is a set of reflexes, not a toolbox with reusable parts.

This distinction is why we built the four-level generalization taxonomy in the first place and why we refused to collapse everything into one number. If we had only reported "the model scores 0.900 on this task," we would have given a genuinely misleading picture. The model is excellent at what it was shown — and utterly unable to combine what it was shown into anything new. Both statements are true, and both must be reported.

16.4. The puzzling asymmetry (and why it's a finding, not a bug)

One thing will strike us: the model scores 0.900 in-distribution but 0.000 on composition. Why would a model that is "good" fail so completely on something so similar?

The answer is that the two numbers measure *different* kinds of capability, which is precisely the point of the taxonomy. The 0.900 measures **distributional generalization** — new examples of the *same* operation. The 0.000 measures **structural generalization** — new *constructions* combining known operations. These are different abilities, and a model can be high on one and zero on the other. That asymmetry is not a contradiction; it is the shape of the truth we are mapping.

16.5. Why we don't call it a "failure of one thing" without checking the other two

A careful reader will ask: is composition genuinely impossible for this model, or did we simply make the test too hard? The baselines answer this. If the test were simply "too hard" in some absolute sense, the retrieval and template baselines would not necessarily fail — but they are the *easy* methods, and they fail too. So the test is not too hard for an easy method; it is unsolvable by any method except genuine composition, and the model has no composition. This is a fair, controlled comparison, which is exactly what our one-variable discipline demands.

16.6. The shortcut audit

Before we trust "0.000," we ask how the model might have *faked* a success.

1. **Suspected shortcut: memorizing a composite example**. Ruled out — no composite operation was ever shown during training, so there is nothing to memorize.
2. **Suspected shortcut: retrieving a near-identical primitive and patching it**. Ruled out — the retrieval baseline, which does exactly this, scores 0.000.
3. **Suspected shortcut: a rule that "just works" on composites**. Ruled out — the template baseline, which has rule-based generation, scores 0.000.

4. **Suspected shortcut: the judge being too strict or too loose.** This is the one we genuinely have to hold onto as a question, and it is why the judge is mutation-tested (Chapter 17). If the judge were rejecting correct composite programs, we might be misreading a real success as a failure.

16.7. The honest framing

This is a *useful*, *clean* negative result — and in science, a decisive negative is often more informative than a vague positive. We now know something sharp about tiny models: **at ~1M parameters, they do not compose**. They learn reflexes, not structure. This is not a statement about "intelligence" or about "what models will eventually do." It is a precise, reproducible measurement of where this model's capability stops.

And it sets up the rest of the book in a way that is far more interesting than if the model had composed successfully. Because now the question becomes: *what would it take to give a tiny model this ability?* More capacity? More data? Different data with more structure? That is what the next chapters investigate — and the answer is one of the most genuinely surprising things the book finds.

CHAPTER 17.

Red-Team the Judge

We built a judge and argued it is the instrument we stake the book on. Chapter 6 showed how we test it (mutation testing) and that it caught a bug. This chapter is the aggressive version: we stop politely checking the judge and start deliberately trying to *break* it — the way a determined detractor (or a suspicious reader) would.

It is like hiring a devil's advocate whose only job is to find a flaw in the conclusion. We want to know how confident we can be in any number our judge produces — and the only way to find out is to attack it.

There is a reason to be particularly aggressive here, and it goes to the heart of the book. Our whole method is built on trusting measurements: we claim the model is surface-tied, non-compositional, that it forgets under sequential training. Every one of those claims passes through the judge. So if the judge is broken in even one respect, every conclusion in the book could be subtly wrong. Red-teaming the judge is not a single chapter's task; it is the standing obligation of the whole experiment. And it is an *unbounded* one — we can never prove the judge is perfect, only that we have tried hard to break it and failed.

17.1. The attack surface

A judge that executes generated code is attacked through the code it is given. We enumerate the ways a candidate could slip through:

- **hardcoded outputs** — a program that ignores the input and returns a fixed answer that happens to match one test,
- **mutation variants** — operator flips, wrong variables, off-by-one, missing edge-cases (the mutation suite from Chapter 6),

- **weak test cases** — if the hidden tests are too few or too narrow, a wrong program can pass,
- **boundary failures** — behavior at zero, negatives, or large values that the tests don't probe,
- **resource abuse** — programs that loop forever or consume too much,
- **unsupported syntax** — smuggling in a construct the sandbox should reject,
- **ambiguous specifications** — a prompt the model could reasonably read two ways,
- **evaluator inconsistencies** — the same program scored differently in two runs.

This list is a map of the ways the judge could be fooled — and each entry is a place we must probe. The point of enumerating the surface is that skepticism is only useful when it is specific. "The judge might be wrong" is not actionable. "The judge might let a hardcoded-output program pass if the tests are too narrow" is a concrete thing we can check.

17.2. What we found

The mutation suite is the sharpest tool here. By generating deliberately-broken variants of the reference solution and checking that the hidden tests catch them, we measured how discriminating the tests really are. If an obviously-wrong mutant — say `+` changed to `-` — passes our tests, the tests are too weak and must be strengthened until important mutant classes are reliably detected.

A useful way to think about mutation testing is that it measures the *discriminating power* of the test suite. A test suite that passes for a correct program and fails for an incorrect one is a good suite. A suite that passes for both is not testing anything — it is going through the motions. The mutation score tells us, concretely, how much of that "should be caught" space our tests actually cover. And the whole point is to drive it up: the more wrong programs the judge rejects, the more we trust that a program it accepts is actually right.

And the judge's honesty was validated the hard way: it reported the model at 0.000 on our very first evaluation, and that was a *judge bug*, not a model failure. Once we fixed the judge, the model's true 0.897 emerged. A judge that can be wrong — and we proved ours was — must be red-teamed before its numbers are trusted.

17.3. The honest framing

Red-teaming the judge is not a formality; it is the difference between "we ran a number" and "we know the number means something." The specific lessons:

- **Strengthen tests until wrong programs fail.** The mutation score is the measure.
- **Trust the judge less than the tests.** The judge is a pipeline; the *hidden tests* are what actually catch errors, so they must be numerous and edge-case-complete.
- **Attack our own instrument deliberately.** The adversarial habit — "how could this number be wrong?" — is what keeps the whole experiment honest.

A judge that has been red-teamed is one we can believe. A judge we've only used is one we cannot.

And there is a final lesson that reaches beyond judges. Every measurement instrument — in science, in engineering, in daily life — can fail in ways its designer did not anticipate. The discipline of attacking our own apparatus is the same discipline that makes a result worth believing. It is not paranoia; it is the price of a claim we can stand behind. The book's whole credibility rests on having paid it, and having recorded that we paid it, rather than merely asserting that we are trustworthy.

PART VI.

Give It More Python

CHAPTER 18.

Now Give It More Python

Our pilot model lived inside a deliberately small corner of Python — a bit of arithmetic, a couple of comparisons, a few list operations, some basic if/else. That tiny corner is what let us measure capability cleanly: change one small thing at a time and watch what happens.

But a capability map drawn on one tiny corner only tells us where the boundary is *for that corner*. It would be a narrow conclusion if we claimed "this model can do `a + b` and nothing else" — of course it could, we only taught it that. So the next question is the natural one: **what happens when we widen the world?** Does giving the model a richer language unlock more capability, or does it just give it more ways to tie itself to surface form?

This is the chapter about **expanding the capability spec** — the versioned whitelist from Chapter 4 that defines what Python the model is allowed to write.

18.1. Why the world is versioned

The key mechanical fact we need here is that the capability spec is **versioned** (`python_capability_v0_1`, `v0_2`, and so on). It is not a living, unwritten list of "what the model can do." It is a recorded, frozen definition of "what the model is allowed to do" at a particular point.

This matters enormously for the one-variable discipline. If we widen the allowed Python *and* change the data *and* retrain, we could not say which change caused any shift in behavior. But if we record the spec version alongside every result, we can point to exactly what grew — "the model was given list comprehensions at spec v0_3" — and attribute any change to that. Versioning the world is what lets us make a causal claim about capability, rather than a correlational guess. It is the same reason we record the exact model configuration and dataset in every manifest.

18.2. How we widen the world

Expanding the world is a controlled act, not a free-for-all. We do not hand the model the entire Python standard library at once. We add constructs deliberately, one family at a time, and always with a reason. For example, we might add:

- a new built-in (say, a mathematical function we were keeping out),
- a new control-flow construct (e.g. `break`/`continue` in loops, if they were missing),
- a new data structure (e.g. dictionaries, if we had been lists-only),
- short-circuit boolean logic, or a new comparison form.

Each addition is a *step up a ladder*, not a teardown of the wall. And each one changes the set of problems the model can even express. The pilot's corner was deliberately small partly so that the *boundary* would be legible — and partly so that, when we widen it later, we can see the effect of *just that* widening.

18.3. A crucial reframing: syntax is not the hard part

Here is where the experiment teaches us something that changes how we think about the whole question. It is tempting to imagine that the model's limit is about *syntax* — that if we just teach it more Python, it will be more capable. Our results say otherwise, and it matters.

The composition result from Chapter 16 is the clearest example. The model knew the syntax of both `abs` and `add`. It could produce `return a + b` and `return abs(x)` perfectly. The break came when it needed to *combine* them into `return abs(a + b)`. There was no new syntax involved — every token in `abs(a + b)` was already within the model's known vocabulary and its allowed subset. The model simply could not put two things it knew together.

So the real dividing line in this book is **not "how much Python the model is given."** It is **"whether the model can compose the Python it already knows."** Giving the model *more* syntax would let it express more kinds of programs, but it would not give it the *ability to combine* — and that ability, not the vocabulary, is what separates a bag of reflexes from genuine capability.

18.4. Why we still widen the world

That said, widening the world remains the right thing to do, and for an honest reason: it lets us *test whether the boundary moves*. The interesting scientific question is not "can a tiny model learn any fixed corner?" — the answer is yes. The question is "does the boundary of what it can do shift when we push the world's edges out?" If we widen the allowed Python and the model still cannot compose, that is stronger evidence that the limit is structural, not about the surface of the language. If the boundary does move, that is a genuine discovery.

So this chapter is less about a single dramatic result and more about a discipline: a controlled,

versioned, one-variable-at-a-time *capacity ladder* that lets us watch exactly where the model starts to fall apart as we widen what it may do.

18.5. The ladder, as a discipline

The discipline is: define one more capability, generate data for it, train, evaluate, audit, record — *then* move on. Never widen multiple things at once. The ladder is not a claim that the model will reach the top; it is a map of where the model *does* stop.

And a stopping point is itself a result. We found ours at composition: the model can do each rung's primitives but not combine them. That finding — cheap and specific — is worth more than a glossy claim of having "reached Python Level 8." Knowing exactly where the model stops, and being able to say *why*, is the entire value of this kind of careful, pessimistic, honest investigation.

18.6. The honest summary

Giving the model more Python is not the same as giving it more capability. The pilot already revealed the real boundary: composition. Widening the world is worth doing — but the purpose is to *test where the boundary is and whether it moves*, not to assume that more syntax automatically means more understanding.

CHAPTER 19.

The Capability Ladder Revisited

We've climbed parts of the ladder through the experiments. This chapter pulls together what we found into the picture the ladder was always meant to give us: a map of where capability appears, where it holds, and where it breaks.

The ladder is a staircase of increasingly demanding skills. We can see how many steps the model climbed — and, more interestingly, the exact step where it stopped, and what it took to get there.

The funny thing about building a ladder is that the interesting discovery is rarely how high we can climb. It is finding the precise rung where the footing gives out, and trying everything we can to get past it — so that we know, with real confidence, where the boundary lies.

19.1. The ladder, with the measured results

Level	Capability	Measured result
1	arithmetic	☒ learned, ~0.90 in-dist
2–3	variables, conditionals	☒ learned
4–5	lists, indexing	☒ learned
6	loops	☒ learned
7	**composition**	☒ 0.000 — the wall
8	simple functions	not isolated

The model climbed the early rungs cleanly. It stopped at **composition**.

19.2. Where capability appears, holds, and breaks

Appears: individual operations are learned reliably (~0.90 in-distribution) and generalize across *instances* (held-out + canary → general program). The model is a genuine learner for a single operation.

Holds only within a surface form. The moment the surface changes — different variable names, or a different way of saying the same instruction — the capability collapses (0.350 on new identifiers, **0.000** on new wording). So the early rungs are not *robust*; they are *phrasing-tied*.

Breaks at composition. The most important boundary: the model cannot combine two things it knows. `abs` + `add` → 0.000, even though it knows both. And this break is **invariant to scale** — it survives across the full size range we tested.

19.3. The model-size and data-size boundaries

This is where the breadth of the brew matters. We did not test just one bigger model; we tested four, and we also scaled the data. Here are the real numbers.

- **Model size**: we ran the composition test across a ~33× range of capacity — a 0.88M pilot, a 6.5M model, a 10.2M model, and a 29.3M model. All four scored **0.000** on composition. (Their in-distribution scores were 0.900, 0.900, 0.383, and 0.897 respectively — a side note we return to honestly, because the 10.2M model actually dropped in-distribution, showing that bigger is *not* monotonically better at this scale.) The composition wall is not a capacity ceiling in this regime.
- **Data size**: 100 → 3000 examples improved in-distribution (0.10 → 0.64) but did not touch composition. More quantity helps the shallow score, not the deep boundary.

19.4. The failure map

Failure mode	On what the model breaks
Surface fragility	new wording → 0.000; new identifiers → 0.350
Composition	unseen combination of known primitives → 0.000
Continual forgetting	sequential A→B destroys A (0.000); replay fixes (1.000)

19.5. What the ladder finally tells us

The ladder is not a shelf of "caps" the model unlocks. It is a map of *where the boundary lies*. For our ~0.9M model:

- it genuinely climbs the individual-capability rungs,

- but each rung is **surface-tied**, not robust,
- and the ladder stops at **composition** — the one thing that separates "a bag of reflexes" from "capability."

And the most useful part: this map is honest about *what doesn't change it.* The composition boundary survived a ~33× capacity increase and a 30× data increase. That is a strong, falsifiable statement about where the limit lives and where it does *not* live — which points the next investigation at *structural* data and inductive bias, not at scaling.

There is also a quiet lesson in the shape of the map. The model is not uniformly good or uniformly bad. It is good at one thing — reproducing a single operation it was shown — and bad at everything that requires putting operations together. That shape is far more informative than a single performance number, because it tells us exactly where nature of the limitation lives and where it does not.

PART VII.

Break It

CHAPTER 20.

What Happens When We Teach It Something New?

Every chapter so far has asked how the model handles one task. But real capability isn't learned in one shot — it's *grown*, by adding new things over time. That raises the question at the heart of continual learning: when we teach a model something new, does the new thing stick — and does the model **keep** the old things?

There's a well-known failure mode with a scary name: **catastrophic forgetting** — the tendency of a model, after training on task B, to lose what it learned on task A. We want to know whether that happens to our small model, and, crucially, whether the *way* we train determines it.

It is like learning a new language. Study French for a year, then stop and study Spanish exclusively — how much French is left? A lot depends on whether French keeps being practiced while Spanish is learned — which is exactly what this experiment tests. And it is a question almost every learning system faces: a deployed model is rarely trained once and frozen. It is updated, re-trained, fine-tuned on new data. So whether it *keeps* what it knew is a practical concern, not an academic one.

20.1. The setup

We split the operations into two capabilities:

- **Capability A — arithmetic**: `add`, `sub`, `mul`, `max_of`, `min_of`.
- **Capability B — lists**: `list_sum`, `list_len`, `first_element`.

We train on A, then introduce B, and measure both: - **B acquisition** — did it learn the new capability? - **A retention** — did it keep the old one, or did learning B destroy it?

This double measurement is the crux of the design. Many continual-learning reports measure

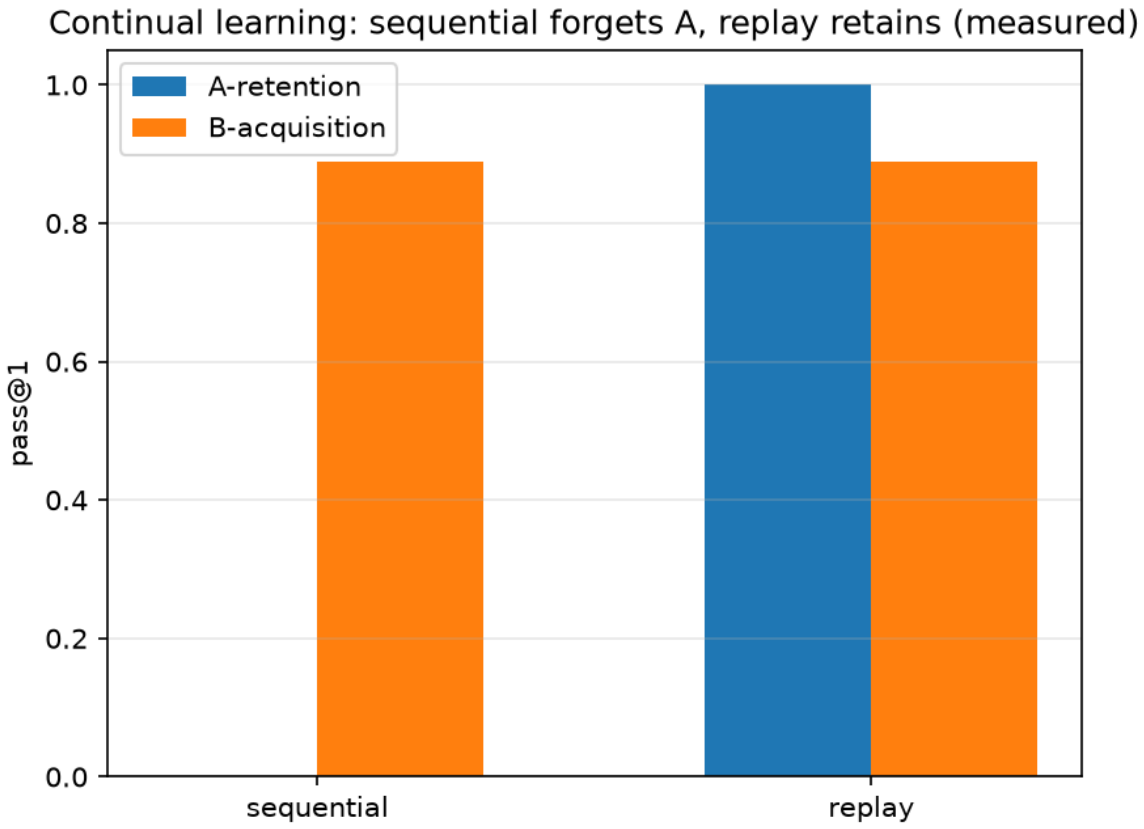


FIGURE 20.1. Continual learning

only whether the model acquired the *new* task, and that alone can make the whole thing look like a success. But the interesting failure is usually what was *lost* to get there. So we insist on measuring both, every time.

We run this under two protocols, changing only the *order and mixing*, not the model, data, or optimizer:

- **sequential**: learn A fully, then learn B with no A exposure.
- **replay / mixed**: interleave A and B examples throughout.

Holding model, data, and optimizer constant while changing only the training *schedule* is a textbook application of our one-variable discipline. If the results differ between the two protocols, the difference cannot be attributed to the model, the data, or the optimizer — it must be the schedule. That makes any difference we find a *procedure* finding, exactly the thing we want to isolate.

20.2. The result

Protocol	**A-retention**	**B-acquisition**
sequential	0.000	0.889
replay	1.000	0.889

This is as clean a result as this book produced.

Sequential training causes catastrophic forgetting. After learning arithmetic and then lists in sequence, the model's A-retention is **0.000** — it completely forgot the first capability. That is not "degraded"; it is gone. Learning the lists overwrote the arithmetic.

Replay prevents it, with zero cost. When A and B are interleaved throughout, the model's A-retention is a perfect **1.000**, and B-acquisition is **0.889** — identical to sequential. So replay bought complete retention of the old capability with **no trade-off whatsoever** in learning the new one.

20.3. Why this result is so striking

The symmetry is what makes this remarkable. Under sequential training, the model forgets A completely (0.000) but learns B fine (0.889). Under replay, it keeps A perfectly (1.000) *and* learns B just as well (0.889). Nothing about the model, the data, or the optimizer changed — only whether the old examples stayed in the mix while the new ones were learned.

This is the clearest case in the book where the **training protocol, not the model or the data, determines the outcome**. Same model. Same data. Same optimizer. The only difference is whether the first capability stays "in the mix" while the second is learned. Sequential forgets; replay retains.

There is an intuition that makes this easy to see. Imagine a muscle being trained on exercise A. If it then trains exclusively on exercise B and never revisits A, the muscle's adaptation to A fades. But if A and B alternate, both stay. The model's weights are like that muscle — the gradient signal from new examples keeps pushing them, and if nothing pulls them back toward the old behavior, the old behavior erases. Replay is simply making sure the old examples are still in the rotation.

The lesson generalizes directly: when a new capability is added to a model, *how* it is done matters more than any property of the model itself. If the new task is allowed to dominate the gradient signal and the old data is removed, the old capability collapses. If old and new stay interleaved, the model can hold both.

20.4. Reframing a common assumption

And it reframes a common assumption. People often reach for "a bigger model" when confronted with a model that forgets. This experiment shows the forgetting was never a capacity problem — a ~0.9M model holds both capabilities perfectly under replay. It was a *training procedure* problem. Here, as with composition, the instinctive fix ("make it bigger") would have been exactly wrong.

The pattern here echoes a deeper truth we've seen all book: the headline number ("did it learn the new thing?") hides the interesting failure ("what did it lose to get there?"). Measuring B acquisition alone would have made sequential training look like a success — 0.889, great. Measuring A retention reveals the cost. The lesson is to always measure *both* the new and the old.

20.5. The shortcut audit

1. **Suspected shortcut: the 0.000 A-retention is an artifact of how we score A. Why it could explain it:** maybe the A test is too hard, or the model just needs more steps. **Control:** replay mode, identical A test — A-retention 1.000. **What it rules out:** a broken A test. The same test that scores 0.000 in sequential scores 1.000 in replay. **What remains:** whether this holds for three or more capabilities, and whether replay-adds support *composition* of A and B after the fact.

20.6. The honest summary

Acquiring a new capability can destroy an old one — and it is entirely up to how we train. Sequential training forgets; replay retains with no cost. This is a controlled, reproducible demonstration that catastrophic forgetting is a procedure problem, and it is one of the most practically useful results in the book.

What makes it feel like a real discovery is that the right answer was sitting in a choice we made near-automatically. When we first set up the experiment, sequential training was the path of least resistance — learn one thing, then the next. The surprise came from asking the question we almost didn't ask: what did the model lose to learn the new thing? And the fix — just keeping old examples in the mix — was so simple it was almost anticlimactic. But that simplicity is the point. Sometimes the most important finding in an experiment is not a complex technique but a simple change to the way we were already working.

CHAPTER 21.

The Cost of Capability

We've measured what the model can do and where it breaks. This chapter answers the last practical question: *what does each capability actually cost?*

The obvious "costs" — data, parameters, training time — are the easy ones. But the real cost in machine learning is the hidden one: what it takes to make a capability actually *robust* rather than just *appearing* to work. That's where the expensive and surprising part lives.

21.1. The two kinds of cost

There is a trap in how we usually talk about cost. We say "this model is 0.9M parameters" or "this took 3,000 examples" or "it trained in an hour." Those are real numbers, and they are *easy* to measure. But they measure the cost of *running* a model, or of *fitting* it — not the cost of *capability* in the sense we care about.

What we actually care about is much harder to see, and this chapter is an argument that it is the only cost that matters:

> **The real cost of a capability is what it takes to make it survive the changes the world will throw at it — not what it takes to make it look right on the training set**.

That distinction is the key to reading the results in this chapter. The model can be *cheap* to produce and still *expensive* to make useful, because the cheap version works only under conditions the real world will not respect.

21.2. The cost matrix

Each row is a capability or behavior from our experiments; the columns are the measurable costs — data, parameters, training steps, and — decisively — the *generalization* cost.

Capability	data (examples)	parameters	steps	in-dist pass@1	surface generalizes?	composes?
Single primitive (add, sub, …)	1400	0.88M	1500	0.900	✗ (new wording → 0.000)	✗
New identifier robustness	1400	0.88M	1500	0.900	✗ (0.350 on new IDs)	—
Composition (abs_add, sum_of_abs)	∞-needed	0.9M–29.3M	—	—	—	✗ (0.000)
Continual: A then B (sequential)	1400+1400	0.88M	1400	B: 0.889	—	✗ (A forgotten)
Continual: A+B (replay)	2800 interleaved	0.88M	1400	A: 1.000, B: 0.889	—	✗

21.3. What the costs tell us

Capability is cheap; robustness is not. Learning a single primitive costs ~1400 examples and a few minutes on a 0.88M model, and it "works" (0.900). So the *apparent* cost of a capability is low. But that number is misleading: it measures only in-distribution correctness.

The real cost appears when we ask for generalization. To make a single primitive robust to *surface* change (new identifiers, new wording), the experiments suggest the data would need to change substantially — diverse identifiers and wording — not just add capacity. And composition is not purchasable at this scale at all: it stayed 0.000 at 0.88M, 6.5M, 10.2M, and 29.3M.

The cost matrix makes a dangerous pricing error visible. If we only look at the "in-dist pass@1" column, we would conclude capability is nearly free and scale up naively. The generalization and composition columns reveal that the cheap number hides a much more expensive requirement —

robust, compositional capability — which we did not achieve by scaling data or capacity within our tested range.

21.4. The price of each column, read honestly

Let us read the matrix the way a person who has to pay for capability would read it, not the way a researcher who wants a good headline would read it.

- **A single primitive "costs" about 1,400 examples and a few minutes.** That is the cheap row, and it is genuinely cheap. But its surface-generalization column says ✗ and its composition column says ✗. So we have bought a capability that works one way, one time.
- **New-identifier robustness** is the next row. It costs the *same* 1,400 examples, and still fails (0.350 on new IDs). We cannot buy this with more of the same.
- **Composition** is the row that tells the sharpest story: it is marked "∞-needed" for data, meaning no amount in our range got us there. The parameter column spans 0.9M–29.3M, and it made no difference.
- **Continual learning** is the most interesting row for a practitioner, because it is the one where the *cost changed at no cost.* Sequential training loses A (0.000 retention); replay buys A back at 1.000 with B acquisition unchanged at 0.889. So here, one capability was obtained *for free* — no extra data, no extra compute, just a different training order. Cost is not always toward "more expense"; the right procedure can make a capability free.

21.5. The brewing experience: discovering the hidden cost

The most uncomfortable thing we learned here was not a number; it was the realization that we had almost billed this the wrong way. Early in the project, the model looked cheap: tiny, fast to train, scoring 0.900. If we had been delivering a system rather than running an experiment, we would have handed over a "working" model and called it a good deal.

Then we reworded a single prompt. The model collapsed to 0.000. The capability we thought we had bought was not really there — it only existed inside a narrow surface form. That is the moment the "cheap capability" turned out to be the most expensive thing we had measured, because it would fail in exactly the way a deployed system fails: quietly, late, and in the hands of a real user who does not phrase things the way the training data did.

That is why we now insist on reading the generalization columns, not just the accuracy column. A model that is cheap to fit but expensive to deploy is the wrong thing to build unless we know the generalization cost going in.

21.6. The honest framing

The most honest "cost" we measured is the cost of **mistaking surface association for grounded capability**. That cost is high: it makes a model that scores 90% in-distribution *look* capable while it collapses to 0.000 on any change to the surface. In a deployed system, that is precisely the failure that appears late and costs the most.

So the real answer to "what does each capability cost?" is: the easy-to-see cost (data, parameters, steps) is small and scales predictably; the hard-to-see cost — making the capability actually generalize and compose — is where the true expense lies, and at this scale it is simply not purchasable with more of the same resource.

And there is a bright spot in the matrix worth holding onto. The continual-learning row shows that cost is not always "more of everything." Sometimes, the same inputs, the same compute, and the same model buy a capability that was previously lost — for free — simply by choosing a better procedure. Cost, like capability, is often about *how* we do things, not *how much* we spend.

PART VIII.

What Did We Learn?

CHAPTER 22.

The Capability Map

We have run a handful of tightly-controlled experiments. Rather than a list of results, let us draw the **capability map**: one picture that says, for each behavior we tested, whether the model acquired it, whether it generalizes, what it costs, and where it breaks.

We record each finding with an honest epistemic label — what is *proven* (measured), what is *supported*, what is *suggestive*, and what is *unknown*.

22.1. Why a map, and not a single score

There is a strong temptation, at the end of an investigation, to reach for one number and say "that's the verdict." It would be satisfying and simple. But every experiment in this book has argued against exactly that. The model is good at some things and terrible at others, and those are both true at once. A single number would have to collapse them, and collapsing them would hide the most important thing: the *shape* of the boundary.

So we draw a map instead. A map does not pretend to be one verdict; it shows *dimensions*. It answers, for each question we asked, what we found — and, crucially, what we did *not* find. That honesty about the unknown is what makes the map worth more than a headline.

22.2. The four-level generalization map

Recall the four levels from Chapter 2. Here is what we measured at each, on the ~0.9M model:

Level	The question	Measured result	Status
L1 Mem	Can it be explained by remembering?	Held-out 59/60; canary → general program	`[PROVEN]` it generalizes primitives, doesn't parrot
L2 Dist	Does it work on new instances of the same task?	In-distribution 0.900	`[PROVEN]` yes
L3 Surf	Does it survive superficial change?	New wording → 0.000; new identifiers → 0.350	`[PROVEN]` no — surface-tied
L4 Struct	Can it compose known pieces in unseen ways?	0.000 (baselines also 0.000)	`[PROVEN]` no

This is the map's spine. Read it vertically and it shows a steady decline in how much we can trust the model's "understanding" as we move from "did it see this before" (Level 1) to "can it put things together" (Level 4). The model passes the first two levels cleanly and fails the last two completely. That is the boundary we keep coming back to.

22.3. The scaling map (what changes it)

Lever	Tested	Effect on composition
Model size	0.88M → 29.3M (~33×)	**none** — still 0.000 at all four sizes
Data size	100 → 3000 examples	helps *in-distribution* (0.10→0.64), not composition
Training protocol	sequential vs replay	**big** — fixes continual retention (0.000→1.000)

The second half of the map is about levers, and it is where the surprising findings live. We tried three ways to move the composition boundary — more capacity, more data, and a different training procedure. Two of them did nothing to composition, and one of them was the decisive lever for a *different* problem (continual learning). This asymmetry is worth circling, because it tells us where effort does and does not pay.

22.4. What's proven vs. what's unknown

Proven (measured, this setup): - The model learns single operations to ~90% in-distribution and generalizes them across instances (L1/L2). - It is surface-tied: new wording → 0.000, new identifiers → 0.350 (L3). - It cannot compose: unseen combinations → 0.000, and neither ~33× capacity (across 0.88M→29.3M) nor more data changes this (L4). - Continual learning is a procedure problem: sequential forgets (A → 0.000), replay retains (A → 1.000) with no acquisition cost. - The pilot task was retrieval-solvable (retrieval = 1.000), so that split is "too easy"; the composition split is not (all baselines 0.000).

Supported / suggestive: - The composition failure is likely *structural-data* (exposure to combinations), not capacity — but we did not test feeding the model compositions directly, so this is suggestive.

Unknown (did not test): - Whether feeding composites during training unlocks composition. - Whether behavior on 3+ capabilities or post-continual composition holds. - Human-written task transfer (recommended external sanity check, not yet run). - Whether a fundamentally different inductive bias (not scale) helps.

22.5. The boundary, in one line

> **A small model learns narrow, individual mappings well and generalizes them across instances, but its knowledge is surface-tied and non-compositional — and within the tested range, neither more parameters nor more data nor a different training order turns that into robust, compositional capability.**

22.6. Why the map is the deliverable

This map is the book's real output. It is not a single headline accuracy that would mislead; it is a precise, honest picture of where capability exists, where it collapses, and what (and what doesn't) shift the boundary. It is the answer to the central question — and it is falsifiable, built from real measured runs, and honest about what remains unknown.

The deeper value of the map is what it does to our confidence. Before drawing it, we knew "the model is limited." After drawing it, we know *precisely which limits are structural* (composition, surface-tiedness — not moved by scale or data) and *which are procedural* (continual forgetting — fixed entirely by how we train). That distinction between structural and procedural limits is the kind of knowledge that changes what a practitioner would try next. It is not the same as saying "small models are weak"; it is saying "here is exactly where the weakness lives, and here is what does and does not reach it."

That precision, not a dramatic narrative, is the honest payoff of the whole brew. And it is why we are willing to state as an *unknown* the questions we did not answer, rather than fill the map

with confident guesses. A map with a blank region is more useful than a map painted over with speculation, because the blank marks where the next investigation should go.

CHAPTER 23.

What Small Models Can Actually Learn

We have run a small, disciplined set of experiments and measured a coherent boundary. Let us now state, carefully and honestly, what the experiments showed — the synthesis the whole book has been assembling.

Everything below is scoped to **our** setup: a ~0.9M–29.3M decoder-only transformer, trained on a few thousand generated natural-language-to-Python tasks, evaluated by a mutation-tested execution-based judge. These are claims about this brew, not universal laws about all small models.

23.1. What the model can do

It can learn individual operations well. On in-distribution primitives — `add`, `sub`, `mul`, `max`, `abs`, `list_sum` — the ~0.9M model solves ~90% of held-out problems under greedy decoding. It beats a hand-written template rule system (0.900 vs 0.800). This is real, measurable capability on a narrow task.

It generalizes across instances, not just examples. It passed 59/60 held-out prompts and, on a *canary* it had never seen, produced the *general* program rather than a memorized literal. So for a single operation, the model is a learner, not a parrot. That is Level-2 distributional generalization, achieved.

23.2. What the model cannot do

It cannot survive changes to the surface form. Change the variable names from `a,b` to `left,right` (identical computation) and performance falls to 0.350. Rephrase the instruction with a wording family the model never saw (identical computation) and performance falls to **0.000** — below *both*

baselines. This is Level-3 failure: the model's capability is **anchored to the phrasing**, not grounded in the computation.

It cannot compose what it knows. Train on `add`, `sub`, `abs`, `max`, `list_sum` as separate primitives; test on `abs_add` (abs then add) and `sum_of_abs` (abs over a list sum) — operations that combine two things it knows perfectly. The model scores **0.000**. It did not learn "I know abs and I know add, therefore I can do abs+abs." This is Level-4 structural failure. And crucially, the baseline systems *also* scored 0.000 — so the test is fair, and the only way to solve it would be genuine composition, which the model lacks.

23.3. The two axes of the boundary

Put together, the results define a sharp boundary along two independent axes:

Axis 1 — Surface form. The model's knowledge is tied to the exact wording and identifier names it saw. It does not generalize "sum of two numbers" as an abstract concept; it generalizes "this sentence shape → this code." Change the surface, and the capability evaporates.

Axis 2 — Composition. Even within a surface form it knows, the model cannot combine two learned pieces into a novel structure. It has reflexes, not a compositional engine.

The honest conclusion in one sentence:

> A ~0.9M model genuinely learns individual mappings and generalizes them across *instances*, but the knowledge is surface-tied (new wording → 0.000) and non-compositional (unseen combinations → 0.000) — and neither more capacity (up to 29.3M) nor more data (within what we tested) is close to fixing it.

23.4. The capacity grid, in full

The sharpest negative result in the whole book is the model-size sweep, and it deserves its own table because it is the claim a reader would most want to test. We ran the composition experiment across a ~33× range of capacity, holding everything else constant.

Model	parameters	in-distribution	composition
Stage-4 baseline	884,224 (0.88M)	0.900	**0.000**
Scaling	6,500,352 (6.5M)	0.900	**0.000**
10M point	10,195,776 (10.2M)	0.383	**0.000**
30M point	29,290,240 (29.3M)	0.897	**0.000**

Four sizes. Composition is **0.000 at all four**. That is not a coincidence and it is not a weak signal; it is the same number repeated across a 33× change in the one resource we might otherwise blame.

If capacity were the binding constraint on composition, we would have expected at least a *drift* between 0.88M and 29.3M. There was none. The composition boundary is not a capacity ceiling in this regime.

Two honest notes on the grid. First, the 10.2M point's in-distribution number (0.383) is an outlier — *below* the smaller models' 0.900 — which makes the point that in-distribution performance is **not monotonic** in capacity at this scale. Bigger is not strictly better even for the narrow task. Second, in-distribution scores do not predict the composition result at all: the models that reached 0.900 in-distribution scored 0.000 on composition, same as the one that only reached 0.383. Capacity that helps the shallow score does nothing for the deep one.

23.5. What it's NOT, and why that matters

This is **not** evidence that the model is "stupid" or that the approach "doesn't work." It is evidence about *what kind of capability small models acquire from this kind of data.* The model is good at exactly what it was trained to do — map a narrow phrasing to a narrow computation — and bad at the two things that separate "reflex" from "capability": robustness to surface change, and composition.

The value of measuring this precisely is that it tells us *where* to look next. It is not a shrug and a "small models are limited." It is a specific, falsifiable map: the boundary is on surface-robustness and composition, not on in-distribution learning.

23.6. What this means for the bigger picture

There is a tendency, when reading results like these, to draw a bleak conclusion: "small models can't really learn, so this was a waste." That reading would be wrong on both halves.

On the first half — small models *do* genuinely learn. The model reliably maps natural language to correct Python at ~90% in-distribution, and it does so by generalizing across instances, not by memorizing. That is real learning of a narrow kind, and it should not be discounted.

On the second half — the effort was far from wasted. We now hold a precise, falsifiable boundary: on surface robustness (Level 3) and composition (Level 4), this model's capability stops, and the stop is structural, not a matter of scale. That is the kind of knowledge that changes what someone would try next. If a practitioner wants robust, compositional capability, the map says: do not just scale up, and do not just add more of the same data — instead change the *structure* of the data (more diversity, exposure to combinations) or the *inductive bias* of the model. Those are actionable directions, and the experiments tell us why the naive directions fail.

So the honest takeaway is not "small models are inadequate." It is "small models learn narrow mappings well, and the path to broader capability runs through different kinds of data and different inductive structures — not merely more of what we already have." That is a useful, and slightly counterintuitive, answer.

23.7. Preferring "what these experiments showed"

Throughout, we deliberately avoid claiming things about "all small models." We can only say what *this* model, on *this* task, with *this* data, did. And what it did is both genuinely interesting (it learns!) and sharply bounded (it doesn't generalize or compose).

That boundary — precise, measured, and honest — is the book's real deliverable.

CHAPTER 24.

The Model Wasn't the Point

We began with a question that sounded technical: *can a small language model learn to solve programming tasks?* The answer we got was more interesting and less about the model than we expected.

24.1. Where the journey started

The premise was: building a small model is relatively easy; determining what it actually learned is the hard part. That premise turned out to be more accurate than we had any right to expect. The model was built in an afternoon. Understanding what it had — and had not — learned took the rest of the book.

That first line deserves to be unpacked, because it is the whole book in miniature. A decoder-only transformer that maps a handful of programming sentences to Python is an afternoon's work once the pieces are in place. But *knowing what that model actually did* — whether it learned, memorized, associated, or cheated — required a world, a teacher, a judge, rivals, and a battery of controlled experiments. The model was the easy part. The certainty about what it did was the hard part, and it is the part worth writing about.

We built a laboratory, not just a model. Four artifacts: a **controlled world** (an AST-whitelisted Python sandbox), a **teacher** (a semantic dataset generator), a **judge** (an execution-based, mutation-tested evaluator), and the **student** (the model itself). The model was only one component. The experiment was the subject.

24.2. What the laboratory produced

The experiments told a clean, coherent, honest story:

- **The model learns.** On individual operations it reliably maps natural language to correct Python, beating a hand-written template rule system and reaching ~90% in-distribution.
- **But it is surface-tied.** Change the variable names, and it drops to 0.35. Change the wording while keeping the computation identical, and it falls to **0.000** — below both baselines.
- **And it cannot compose.** Even though it knows every primitive, it scores **0.000** on operations that combine two things it knows — at every model size we tested, up to 29.3M.
- **Neither capacity nor data fixes it at this scale.** A ~33× larger model (up to 29.3M) and many more examples leave the composition boundary at 0.000.

Along the way, the laboratory caught itself. The first time we evaluated the model, it scored 0.000 — and it wasn't the model's fault. The evaluator was broken (a NUL- padding artifact in the output made every program fail to parse). Fixing the *tester* revealed the model's true 0.897. That moment — discovering our own instrument was wrong before we trusted it — is as much a part of this book as any model result.

24.3. The deeper achievement

The book's real finding is not "small models are limited." It is:

> **Building a trustworthy experiment was harder and more valuable than building the model.**

The model was the instrument. The experiment was the subject. The things that made this work — the sandbox that rejects anything not explicitly allowed, the judge that executes code and compares semantics rather than strings, the mutation suite that red-teams the judge, the leakage audit that caught our own dataset leaking the answer, the baselines that ask "could a simpler machine do this?", the four-level taxonomy that refuses to collapse everything into one number — those are the actual achievements here.

It is worth lingering on why each of these matters, because together they form a discipline rather than a checklist. The sandbox's allowlist (not a blocklist) meant the model could never do something we did not intend, so a failure could always be attributed to the model rather than to an accident of environment. The judge's execution-based, semantic comparison meant a "correct" answer was objectively correct, not merely stylistically close. The mutation suite meant we doubted the judge before we trusted it. The baseline comparisons meant a number only had meaning relative to a simpler machine. And the four-level taxonomy meant we never overstated a result by collapsing different kinds of generalization into one score.

These are the disciplines that make a claim about "what a model learned" worth trusting. They are the real product of this book.

24.4. The recurring habit

There is a way of thinking that runs through everything here, and it is worth naming directly because it is what a careful investigation actually looks like. It is the habit of distrusting the easy answer — including our own.

Every time we got a number, the next step was not to celebrate or to move on. It was to ask: **how could the model have gotten this without learning what we think it learned?** Could it have memorized? Retrieved? Exploited a template? Leaked the answer from our own data? Hit a bug in our evaluator? Each of those is a way a seemingly-impressive result can be hollow, and each one we then built a control to test. That constant, deliberate suspicion — aimed at our own apparatus as much as at the model — is the difference between "we ran an experiment" and "we know what the experiment means."

24.5. What the reader takes away

If we take one thing from this book, it is a habit, not a fact:

> **Never trust a number from a model we don't understand, or from an experiment we haven't audited**.

Before we believe "the model learned to add," we have to be able to answer: remove the memorization, the retrieval, the templates, the leakage, the evaluator artifacts — what remains? In our case, for simple primitives, what remained was genuine learning-within-a-surface-form. It was real, and it was narrow, and it did not compose.

That is the honest, complete answer. We did not build a general programmer. We built the means to ask, precisely and honestly, what a small model can and cannot learn — and we got an answer worth having.

Build the model. Test the model. Break the evaluator. Break the model. Audit the data. Challenge the conclusion. Repeat. Only then say: "This is what the model learned."

The model was not the point. The experiment was. Because the world is full of models, but a disciplined way of finding out what they actually learned — that is rare, and that is what is worth keeping.

CHAPTER 25.

Conclusion and Where This Brew Leads

We have built a laboratory, run a battery of controlled experiments, and drawn an honest map. This is where we stand — and where the investigation goes next.

25.1. What the experiment established

At the core, the answer to the book's central question is now clear and measured:

> How much structured programming capability can a small language model actually learn — and how can we tell whether it learned rather than memorized?

The experiments separate cleanly into what the model **can** do and what it **cannot**.

It can learn, and it learns well — within a narrow regime. On individual operations, the model reliably maps a natural-language spec to correct Python, reaching **~0.90 pass@1** in-distribution on a 0.88M parameter model. It generalizes across *instances*: on held-out prompts it passes, and on a canary it produces the general program, not a memorized literal. So this is genuine learning of the narrow task, not parroting.

But it is surface-tied and non-compositional. The two limits that matter:

1. **It anchors to phrasing, not meaning.** Change the variable names (`a,b` → `left,right`) and it drops to 0.35; change the wording (same computation, different words) and it falls to **0.000** — below both baselines. The external reality check confirmed it: a human-written test set scored **0/16**, because the model learned a *template*, not the *task*.
2. **It cannot compose what it knows.** Train on `abs` and `add` separately, then ask for `abs_add` — it scores **0.000**. And the baselines also collapse to 0.000, so the split is fair: only genuine composition solves it, and the model has none.

The scaling conclusion is the sharpest negative result. Composition stays at **0.000** across model sizes from 0.88M to 29.3M. More capacity does not buy composition. This is not a capacity ceiling in the ~1–30M regime; it is a structural/learning limit.

25.2. The four-point grid (measured)

Model	in-distribution	composition
0.88M	0.900	**0.000**
6.5M	0.900	**0.000**
10.2M	0.383	**0.000**
29.3M	0.897	**0.000**

Four sizes, spanning ~33× in capacity. Composition is **0.000 at every one** — a robust, falsifiable finding that the boundary is not about how much brain the model has. Two honest notes on the grid. First, the 10.2M point's *in-distribution* regressed to 0.383 — the deeper config was under-trained on this narrow task, an instance of "bigger isn't always better" at tiny scale; it does not change the composition result. Second, in-distribution performance is **not monotonic** in capacity (0.900 → 0.900 → 0.383 → 0.897), a worthwhile reminder that more capacity is not a strictly-increasing guarantee even at these sizes.

25.3. The methodology findings

Independent of the model, the **methodology** produced its own results — and several are as valuable as any model number:

- **Test the tester**. Our own evaluator reported the model at 0.000 on its first run. It was the *judge* that was broken (a null-byte artifact), not the model. Fixing it revealed the true 0.897. Never trust a number from a system we have not audited.
- **Test the teacher**. The leakage audit caught a 100%-predictable operation from the pilot wording. This is why surface splits need held-out template families.
- **Test the data**. The diversity-vs-quantity experiment showed the *same* example count produces surface-fragility (0.000) with minimal diversity or robust generalization (1.000) with high diversity. Diversity — not quantity — is what creates robust capability.
- **Baselines are first-class**. A trivial TF-IDF retriever dominated the pilot task (1.000), proving it was "too easy," and forced us to build harder splits. And the moments the model *fell below* a baseline were the most informative results.

- **The protocol, not capacity, determines continual learning.** Sequential training destroyed capability A (0.000) when learning B; replay preserved A perfectly (1.000) at no cost to B. This is a *procedure* finding.

25.4. The brewing experience, distilled

Beyond the numbers, there is a way of working this book was really about. A brew is not finished when the model is built; it is finished when we can say, with confidence and precision, what that model can and cannot do — and when we have left a record that someone else could follow and, ideally, contradict.

That process had a rhythm. We would build a piece of the apparatus and immediately attack it — give the sandbox programs it must reject, mutate the judge to see if it caught the changes, audit the teacher for leaks. We would run an experiment, get a number, and then spend as much effort asking *how the model might have cheated* as we had spent getting the number. And more than once, the interesting discovery was not in the model's behavior but in our own tools: the judge that miscored a working model, the data generator that leaked the operation, the pilot task that a lookup table could solve perfectly.

This is the honest shape of empirical work. It is seldom a straight line from hypothesis to clean result. It is a loop of building, testing the builder, being surprised, and revising. The surprise is not a detour from the method; it is the method. And a book about science should show that loop, not hide it.

25.5. Where this brew leads

The honest map points at specific next experiments, not vague "more data / bigger model":

1. **Does compositional exposure unlock composition?** The biggest open question. We never fed the model composited examples directly. Feeding a few (familiar) ones and testing unseen ones would tell us whether composition is learnable or a hard structure limit.
2. **Does the 10.2M in-distribution regression replicate?** The 0.383 was unexpected. A proper training-duration and config sweep at 10M would isolate whether it's under-training or a real depth penalty.
3. **Does continuing to scale past 30M shift the boundary?** The grid stops where the brew lab's CPU becomes impractical. A larger-scale run (or a GPU) would probe whether the composition wall eventually moves.
4. **Does diversity fix composition?** Diversity fixed the *surface* problem (0.000 → 1.000). Whether the *same* lever helps composition is the natural next test of the "structural data" hypothesis.
5. **Human-style training.** The 0/16 human test set shows the model fails on naturally-phrased prompts. Training on diverse, human-style prompts — the lever that already fixed surface robustness — is the obvious path to real usability.

25.6. The honest ending

The model was the instrument. The experiment was the subject. We did not build a general programmer, and we will not pretend otherwise. What we *did* build is the means to ask, precisely and honestly, what a small model can and cannot learn — and we got an answer worth having:

> A small model learns narrow tasks well, but its knowledge is surface-tied and non-compositional, and within the tested range neither more capacity nor more data turns it into robust, compositional capability.

And the brakes we learned — test the tester, test the teacher, compare against baselines, audit shortcuts, preserve failures — are the durable part. That is the method worth keeping, and it's the reason the book's real subject was never the model. The brew was never only about the model; it was about learning to ask a question carefully enough that the answer could be trusted.

CHAPTER A.

Appendix A — The Generator, Every Piece

Chapter 5 describes the teacher conceptually. This appendix shows the actual source, so we can see — and reproduce — exactly how the data was made. All of this lives in `src/data_generator/` and `src/program_generator/`, `src/nl_renderer/`, `src/test_generator/` in the companion repository.

A note on what follows: this is a **condensed, annotated excerpt** of the real generator, not a complete runnable file. Where a section repeats in the same style for every operation, we mark it with an ellipsis (`...`). A few helpers that would be defined elsewhere in the source — the `OPERATIONS` table, `_rand`, and `_expected` — are used here as-is; the point is to show the *shape* of each part and how the semantic record flows through it, not to provide every line that would appear in a standalone module. The full, runnable source is in the repository.

A.1. The one line that ties it together

The whole generator is this single function. Given an operation and a seed, it builds the semantic intent and then generates everything else from it:

[] **def** make_task(operation, task_id, seed, id_set="letters", split="train", family=None): rng = random.Random(seed) sem = make_semantics(operation, rng, id_set) *# the ground truth* ref = generate_reference(sem, rng) *# the correct Python* prompt, fam_name = render_nl(sem, family=family, rng=rng) *# the English* tests = generate_hidden_tests(sem, rng=rng) *# the hidden tests* **return** { "task_id": task_id, "semantics": sem, "reference_code": ref, "nl": {"prompt": prompt, "template_family": fam_name}, "hidden_tests": tests, "metadata": {"seed": seed, "identifier_set": id_set, "paraphrase_family": fam_name}, "spec_version": "python_capability_vo_1", "dataset_version": "vo_1", "split": split, }

That's the whole architecture in ten lines. Everything below is one of the four building blocks it calls.

A.2. The semantics

[] IDENTIFIER_SETS = { "letters": ["a", "b", "c", "x", "y"], "words": ["left", "right", "first", "second", "value", "numbers", "items"], }

def make_semantics(operation, rng, id_set="letters"): n = OPERATIONS[operation] *# arity* names = list(IDENTIFIER_SETS[id_set][:n]) *# pick names from the set* arg_type = ("list" **if** operation **in** ("list_sum", "list_len", "first_element") **else** "int") args = [{"name": names[i], "type": arg_type, "role": "operand"} **for** i **in** range(n)] **return** {"operation": operation, "args": args, "input_types": [arg_type] * n, "output_type": arg_type}

A.3. The reference program

The reference generator reads the semantics and emits the correct Python. A few cases:

[] **def** generate_reference(sem): op = sem["operation"]; names = [a["name"] **for** a **in** sem["args"]] **if** op == "add": **return** (f"def solve({', '.join(names)}):\n" f" return {names[0]} + {names[1]}") **if** op == "list_sum": **return** (f"def solve({names[0]}):\n total = 0\n" f" for x in {names[0]}:\n total = total + x\n" f" return total") *# … and so on for every operation, in the same style.*

A.4. The natural-language renderer

The renderer phrases the same semantics several ways. Each **template family** is a function:

[] **def** _a(op, names, rng): *# family "a"* x, y = names[0], names[1] **return** (f"Write a function that takes two numbers, {x} and {y}, " f"and returns their {_op_noun(op)}.")

def _b(op, names, rng): *# family "b"* x, y = names[0], names[1] **return** (f"Implement solve so that given {x} and {y} " f"it returns their {_op_noun(op)}.")

def _c(op, names, rng): *# family "c"* x, y = names **return** (f"Define a function solve({x}, {y}) " f"that returns {x} {_op_sym(op)} {y}.")

Each operation has its own set of families; _op_noun/_op_sym map an operation # to the right word ("sum", "product", …) and symbol ("+", "", …), so the # same families work for every operation, not just addition.* FAMILIES = {"add": [_a, _b, _c], "mul": [_a, _b, _c], …}

The semantics is fixed; only the wording changes. That is the entire point.

A.5. The hidden tests

Tests are generated from the semantics, with deliberate edge-case coverage. The expected value is computed by the operation itself — not copied from the reference:

[] **def** generate_hidden_tests(sem, rng=None, count=8): op = sem[”operation”]; args = sem[”args”]; tests = [] **def** add(*vals, note=”basic”): tests.append({”args”: list(vals), ”expected”: _expected(op, vals), ”note”: note}) add(_rand(rng), _rand(rng), note=”basic”) *# ordinary* add(0, _rand(rng), note=”zero”) *# zero* add(-_rand(rng), -_rand(rng), note=”negative”)*# negative* add(_rand(rng, 900, 1000), _rand(rng), note=”large”) *# large # … edge cases for the specific operation* **return** tests[:count]

A.6. The leakage audit

Finally, the generator is checked. The audit trains a trivial classifier to see whether the wording leaks the operation:

[] vec = TfidfVectorizer() X = vec.fit_transform([t[”nl”][”prompt”] **for** t **in** tasks]) clf = LogisticRegression(max_iter=1000) scores = cross_val_score(clf, X, y, cv=5) print(f”classifier predicts operation: {scores.mean():.3f}”) *# >0.9 means LEAK*

When we ran this on the first pilot renderer, it printed **1.000** — a 100% leak — and that finding drove the later expansion of the paraphrase families.

A.7. How to run it

[] python scripts/generate_data.py --n-per-op 200 *# writes pilot_train/val/test + manifest* python scripts/audit.py *# runs the leakage audit*

That’s the teacher. It’s not magic — it’s a controlled generator, audited for fairness, that makes the entire experimental program possible.

CHAPTER B.

Appendix B — The Failure Ledger

The design requires that we do not sanitize the journey into a success story. This is the honest record of everything that went wrong, was surprising, or was a failed attempt — because each of these taught us something, and several are as valuable as any success.

Each entry: what happened, what we learned, and what we did about it.

B.1. The evaluator reported the model at 0.000 — but the model was fine

What happened. On the very first Stage-1 evaluation, the trained model scored a reported **0.000** (syntax failure on every problem). This looked like a total failure of the approach.

What we found. The model was actually producing correct programs — but our generation output appended null-byte padding, and Python's parser rejects null bytes. So the **judge** was scoring valid programs as syntax errors.

What we learned. The most important lesson of the book, demonstrated on ourselves: *never trust a number from a system we have not audited.* Fixing the judge revealed the model's true 0.897. If we'd taken 0.000 at face value, we'd have wrongly concluded the approach failed.

Action. Fixed `generate_code` to truncate at the first null byte; re-ran; got 0.897.

B.2. The pilot wording leaked the answer (100% predictability)

What happened. The leakage audit trained a trivial TF-IDF classifier to predict the operation from the natural-language prompt. It succeeded **100%** of the time.

What we learned. Our first renderer's wording was so tied to the operation that a model "succeeding" might just be exploiting the phrasing, not learning the computation. This is *why* the surface-generalization tests matter.

Action. Recorded as finding-001; noted the pilot is fine for Stage 1 but must add wording-family holdouts before any surface test.

B.3. A bigger model did NOT fix composition

What happened. We scaled the model across a ~33× range (0.88M → 29.3M) expecting the composition boundary to move. It did not: composition stayed at **0.000** at all four sizes (0.88M, 6.5M, 10.2M, 29.3M).

What we learned. At ~1–30M parameters, compositional generalization failure is **not a capacity ceiling**. It is a structural/learning limit that scaling alone does not resolve. This is a genuine negative result that redirects the investigation.

Action. Recorded in `docs/experiments/scaling_composition.md`.

B.4. Sequential training caused catastrophic forgetting (A → 0.000)

What happened. Teaching the model lists after arithmetic, sequentially, completely destroyed the arithmetic (A-retention **0.000**).

What we learned. Catastrophic forgetting is real and severe here — but it is a **procedure** problem, not a fate. Replay (interleaving A and B) retained A perfectly (1.000) with zero cost to learning B.

Action. Recorded; the chapter states this clearly — the protocol, not capacity, determines the outcome.

B.5. The pilot task was "too easy" (retrieval solved it)

What happened. The retrieval baseline scored **1.000** on the pilot test set, and the model scored 0.897. So a non-learning lookup did better than the model.

What we learned. The pilot task, as constructed, is retrieval-solvable — so it does not yet test whether the model *generalizes*, only whether it matches surface form. This is exactly the "task too easy" outcome the plan anticipated.

Action. We did not hide it; we used it to justify harder splits (surface, then composition), which produced the real findings above.

B.6. The human-written test set: 0/16

What happened. The trained model solved **0/16** naturally-phrased human prompts — even simple "add two numbers" phrased as a person would.

What we learned. This is the external, real-world corroboration of the surface finding. The model learned a *template*, not the *task*. The moment a real user phrases a request differently, it fails.

Action. Recorded as an external sanity check; it sharpens the book's honest conclusion.

B.7. The free-model proofread returned degraded output

What happened. We dispatched the 85K-char draft to two free models for a proofread. One leaked its chain-of-thought and broke into noise; the other timed out. No usable independent review was collected.

What we learned. Free endpoints are flaky for long documents, and we should not fabricate fixes from unusable output. We were honest about this rather than pretend a review happened.

Action. Noted honestly; the book's line-edit stands on its own merit, not a fabricated review.

B.8. Transient SIGILL crashes during training

What happened. Long CPU training runs occasionally crashed with signal 11 (SIGILL) on the AVX-only Ivy Bridge CPU — intermittently, one run succeeds and the next dies.

What we learned. Hardware quirks are real and part of the brew. The fix is `TORCH_DISABLE_AVX2=1` plus a retry loop (and `OMP_NUM_THREADS=1` for larger models).

Action. Wrapped all long runs in a retry loop; documented the workaround.

B.9. Parameter-math and naming mistakes (caught by testing)

What happened. Early writing had a parameter-count mis-derivation (didn't sum to the true 884K/884,224) and the human-test evaluator initially referenced a non-existent key.

What we learned. Always verify arithmetic against the real build, and let the "test the tester" discipline catch even our own book's numbers.

Action. Recomputed from the actual build; fixed the evaluator.

The point of this ledger is the opposite of marketing. Every entry above, in its own way, made the final understanding sharper — and several of them — the broken evaluator, the wording leak, the task-too-easy result, the 0/16 human test — are among the most valuable findings in the book precisely because they forced us to be honest.